

MATH3301

Graph Theory and Design Theory

Graph Theory

2025

Acknowledgements

In addition to the other references cited in the text, extensive use of three books in particular has been made in preparing these notes. These are *Introduction to Graph Theory* by Douglas West [22], *Graphs and Digraphs* by Gary Chartrand, Linda Lesniak, and Ping Zhang [3], and *Four Colours Suffice: How the map problem was solved* by Robin Wilson [25]. Many of the included proofs are adapted from proofs in these books. Also, several factual claims made here are derived directly from these books, sometimes without direct reference to them. Some of the figures are reproduced from images that are freely available on Wikimedia Commons [24].

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Chapter 1

Definitions and Preliminaries

Basic Definitions:

A **graph** G consists of a finite non-empty set $V(G)$ of **vertices**, a finite set $E(G)$ of **edges**, and an incidence relation that associates two, not necessarily distinct, vertices to each edge. The vertices associated with an edge are called its **endpoints**, and the endpoints of an edge are said to be **joined** by the edge. Two vertices are **adjacent** if they are endpoints of some edge, and two edges are **adjacent** if they share a common endpoint. A vertex and an edge are **incident** if the vertex is an endpoint of the edge. The vertices adjacent to a given vertex are called its **neighbours**, and the set of neighbours of a vertex v in a graph G is denoted $N_G(v)$, or just $N(v)$ if G is understood.

An edge whose endpoints are the same is called a **loop**, and a set of edges (more than one edge) with the same endpoints is called a **multiple edge**. A graph is **simple** if it has no loops and no multiple edges. An edge with endpoints u and v may be denoted by uv .

Order, Size, Subgraphs:

The number of vertices in a graph G is denoted by $n(G)$ and called the **order** of the graph. A graph of order n is called an **n -vertex graph**. The number of edges in a graph G is denoted by $e(G)$ and called the **size** of the graph.

A graph H is a **subgraph** of G if $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$ (and each edge of H has the same endpoints in H and G). A subgraph H of a graph G is **spanning** if $V(H) = V(G)$. If G is a graph and $U \subseteq V(G)$ (with $U \neq \emptyset$), then the subgraph of G **induced** by U has vertex set U and contains each edge of G that has both endpoints in U . If G is a graph and $S \subseteq E(G)$, then the subgraph of G **induced** by S has the endpoints of the edges of S as its vertex set and has S as its edge set.

Two simple graphs G_1 and G_2 are **isomorphic**, denoted $G_1 \cong G_2$, if there exists a bijection $f : V(G_1) \rightarrow V(G_2)$ such that $uv \in E(G_1)$ if and only if $f(u)f(v) \in E(G_2)$. If G and H are graphs and G has no subgraph isomorphic to H , then G is **H -free**.

Cliques, Independent Sets, Degree:

A set of pairwise adjacent vertices is called a **clique**, and a set of pairwise non-adjacent vertices

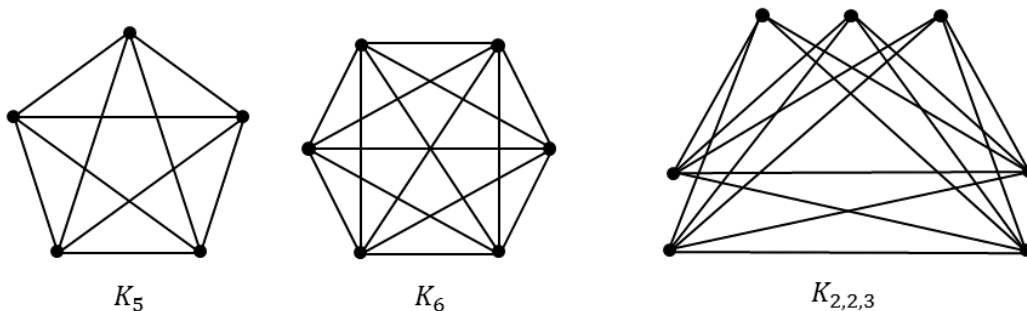
is called an **independent set**. A clique consisting of m vertices is an **m -clique**. The **degree** of a vertex v in a graph G , denoted $\deg_G(v)$ or just $\deg(v)$ if the graph G is clear from the context, is the number of edges of G that are incident on v plus the number of loops on v (so each loop on v contributes 2 to the degree of v). A graph in which every vertex has the same degree is **regular**, and a regular graph of degree k is **k -regular**. The **minimum degree** of a graph G , denoted $\delta(G)$, is defined by $\delta(G) = \min\{\deg_G(x) : x \in V(G)\}$, and the **maximum degree** of a graph G , denoted $\Delta(G)$, is defined by $\Delta(G) = \max\{\deg_G(x) : x \in V(G)\}$.

Theorem 1.0.1. In any graph G ,

$$\sum_{x \in V(G)} \deg_G(x) = 2e(G).$$

Complete and Complete k -partite Graphs:

A graph is **complete** if any two distinct vertices are adjacent, and the simple complete graph of order n is denoted by K_n . A graph G is **bipartite** if $V(G)$ can be partitioned into two independent sets, and is **k -partite** if $V(G)$ can be partitioned into k independent sets, called **parts**. For $k \geq 2$, a simple k -partite graph in which any two vertices from distinct parts are adjacent is a **complete k -partite graph**. The complete k -partite graph with parts of cardinalities $m_1, m_2, \dots, m_k$ is denoted by $K_{m_1, m_2, \dots, m_k}$.



Paths and Cycles:

A **path** is a graph with distinct vertices $v_1, v_2, \dots, v_k$, $k \geq 1$, and edges $v_1v_2, v_2v_3, \dots, v_{k-1}v_k$, and is denoted by $[v_1, v_2, \dots, v_k]$. The **length** of a path is its size (number of edges), which is one less than its order (number of vertices). A path of length m is called an **m -path**, and is denoted by P_{m+1} . The vertices v_1 and v_k are the **endpoints** of the path $[v_1, v_2, \dots, v_k]$ and a path with endpoints x and y is called an **x, y -path**. The vertices of a path which are not its endpoints are called **internal vertices**. A **Hamilton path** is a subgraph that is a spanning path.

A **cycle** is a graph with distinct vertices $v_1, v_2, \dots, v_k$, $k \geq 1$, and edges $v_1v_2, v_2v_3, \dots, v_{k-1}v_k, v_kv_1$, and is denoted by $(v_1, v_2, \dots, v_k)$. The **length** of a cycle is its order, which equals its size. A cycle of length m is called an **m -cycle**, and is denoted by C_m . A 3-cycle is a **triangle**. A **Hamilton cycle** is a subgraph that is a spanning cycle, and a graph that has a Hamilton cycle is **Hamiltonian**.

Distance, Diameter and Girth:

The **distance** between vertices x and y in a graph G is denoted by $d(x, y) = d_G(x, y)$ and defined as the length of the shortest path from x to y . If there is no x, y -path in G , then $d(x, y)$ is infinite. The **diameter** of a graph G , denoted $\text{diam}(G)$ is the largest distance between two vertices of G , that is, $\text{diam}(G) = \max\{d(x, y) : x, y \in V(G)\}$. The **girth** of a graph that has a cycle is the length of its shortest cycle, and a graph with no cycles has infinite girth.

New Graphs from Old:

Let G be a graph let $x \in V(G)$, $e \in E(G)$, $U \subset V(G)$, and $S \subseteq E(G)$. Then $G - U$ is the graph obtained from G by deleting the vertices in U from $V(G)$ and deleting any edges that have endpoints in U from $E(G)$, and $G - S$ is the graph obtained from G by deleting the edges in S from $E(G)$. The notation $G - x$ is used for the graph $G - \{x\}$ and the notation $G - e$ is used for the graph $G - \{e\}$. Also, $G + xy$ is the graph with vertex set $V(G)$ and edge set $E(G) \cup \{xy\}$.

The **complement** of a simple graph G , denoted $\overline{G}$, is the graph with the same vertex set as G and where distinct vertices are adjacent in $\overline{G}$ if and only if they are non-adjacent in G . The **line graph** of a simple graph G , denoted $L(G)$, is the graph whose vertices are the edges of G and where there is an edge e_1e_2 in $L(G)$ if and only if e_1 and e_2 are adjacent edges in G .

Walks, Trails, Paths and Connectivity:

A **walk** in a graph G is a list $v_0, e_1, v_1, e_2, v_2, \dots, e_k, v_k$ of vertices $v_0, v_1, \dots, v_k$ and edges $e_1, e_2, \dots, e_k$ such that for $i = 1, 2, \dots, k$, the edge e_i has endpoints v_{i-1} and v_i . In a simple graph, a walk may be defined by just listing the vertices (because this defines the edges of the walk). The first and last vertices of a walk are called its **origin** and **terminus** respectively, and collectively they are called its **endpoints**. A walk with origin x and terminus y is called an **x, y -walk**. The **length** of a walk is the number of edges it contains. A walk is **closed** if its endpoints are equal, and otherwise it is **open**.

A walk with no repeated edges is a **trail**, and a walk with no repeated vertices is a **path**. Note that the word “path” is used in two technically different ways in graph theory. The graph $[v_1, v_2, \dots, v_k]$ which has distinct vertices $v_1, v_2, \dots, v_k$ and edges $v_1v_2, v_2v_3, \dots, v_{k-1}v_k$, and the walk $v_0, v_0v_1, v_1, v_1v_2, v_2, \dots, v_{k-1}v_k, v_k$ where $v_1, v_2, \dots, v_k$ are distinct are both called paths. A trail with origin x and terminus y is called an **x, y -trail**, and a path with origin x and terminus y is called an **x, y -path**.

A graph is **connected** if there is an x, y -path for all vertices x and y , otherwise the graph is **disconnected**. The maximal connected subgraphs of a graph are called **components**; H is a maximal connected subgraph of a graph G if H is connected and the only connected subgraph of G having H as a subgraph is H itself. A connected graph with no cycles is a **tree**. A **bridge** in a graph G is an edge e such that $G - e$ has more components than G .

Factors, Factorisations, and Colourings:

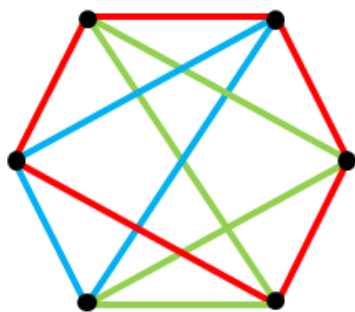
A **decomposition** of a graph G is a set of subgraphs whose edge sets partition the edge set of G . A spanning subgraph is called a **factor**, a k -regular factor is called a **k -factor**, and a decomposition

into k -factors is called a **k -factorisation**. A 1-factor is sometimes called a **perfect matching**.

Theorem 1.0.2. A graph can be decomposed into cycles if and only if every vertex has even degree.

Proof The result holds trivially if there are no edges. Any cycle decomposition partitions the edges at each vertex into pairs (and loops). Thus, if there is a decomposition of a graph into cycles, then every vertex has even degree.

For the converse, we first observe that any nonempty graph in which each vertex has even degree contains a cycle. This is easily seen because any walk starting at a vertex of nonzero degree can continue without repeating edges until a repeated vertex is encountered, in which case a cycle is formed. If the edges of a cycle are removed from a graph G , then the degree in G of each vertex of the cycle is reduced by 2. So if we start with a graph in which every vertex has even degree, then we can greedily remove cycles until a decomposition is obtained. $\square$



A decomposition of a graph into a 3-cycle (blue), and 4-cycle (green) and a 5-cycle (red).

A **colouring** of a graph is an assignment of colours to its vertices, and a colouring with k or fewer colours is a **k -colouring**. A k -colouring is **proper** if adjacent vertices are assigned distinct colours, and a graph is **k -colourable** if it has a proper k -colouring. The term **(vertex) colour class c** refers to the set of vertices of colour c .

An **edge colouring** of a graph is an assignment of colours to its edges, and an edge colouring with k or fewer colours is a **k -edge colouring**. A k -edge colouring is **proper** if adjacent edges are assigned distinct colours, and a graph is **k -edge-colourable** if it has a proper k -edge colouring. The term **(edge) colour class c** refers to the set of edges of colour c . For edge colourings, we sometimes use the term **colour class c** to mean the subgraph induced by colour class c .

If a graph G has a proper k -edge colouring, then it is clear that $k \geq \Delta(G)$. A theorem of König [10] says that we have equality in the case of bipartite graphs, and Vizing's Theorem [20] says that we never need more than $\Delta(G) + 1$ colours for a proper edge colouring of a simple graph G .

Theorem 1.0.3. (König, 1916) Every bipartite graph G is $\Delta(G)$ -edge colourable

Theorem 1.0.4. (Vizing's Theorem) Every simple graph G is $(\Delta(G) + 1)$ -edge colourable.

Digraphs:

A **directed graph**, or **digraph**, is defined similarly to a graph, except that the incidence relation associates an ordered pair (x, y) of vertices to each edge. The first vertex of the ordered pair is the **tail**

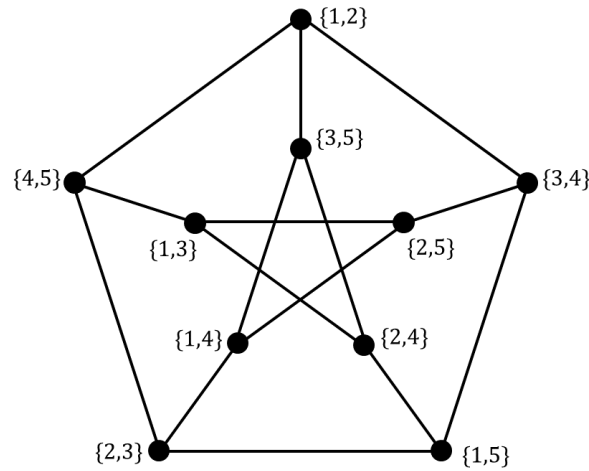
of the edge, and the second is the **head**, and the edge xy is said to be from x to y (whereas the edge yx is from y to x). Many of the above definitions for undirected graphs extend in an obvious/natural way to digraphs. We mention a few additional definitions that are needed and points that require clarification in the case of digraphs.

In a digraph G , the **out-neighbourhood** of a vertex x , denoted $N^+(x)$, is defined by $N^+(x) = \{y : xy \in E(G)\}$, and the **in-neighbourhood** of a vertex x , denoted $N^-(x)$, is defined by $N^-(x) = \{y : yx \in E(G)\}$. The **outdegree** of x , denoted $\deg_G^+(x)$ or just $\deg^+(x)$, is the number of edges of G having x as their tail and the **indegree** of x , denoted $\deg_G^-(x)$ or just $\deg^-(x)$, is the number of edges having x as their head.

The **line digraph** of a simple digraph G , denoted $L(G)$, is the digraph whose vertices are the edges of G and there is an edge e_1e_2 in $L(G)$ if and only if the head of e_1 is the tail of e_2 .

The Petersen Graph:

Definition 1.0.5. The **Petersen graph** has vertex set consisting of the 2-element subsets of the set $\{1, 2, 3, 4, 5\}$ and $\{a, b\}$ is adjacent to $\{c, d\}$ if and only if $\{a, b\} \cap \{c, d\} = \emptyset$.



Theorem 1.0.6. The Petersen graph P has the following properties.

1. P is a 10-vertex 3-regular graph;
2. P has no 3-cycles and no 4-cycles;
3. P has girth 5;
4. Any two non-adjacent vertices of P have a unique common neighbour;
5. P has no Hamilton cycle;
6. P has no proper 3-edge colouring;

Proof The Petersen graph P has $\binom{5}{2} = 10$ vertices. For any vertex $\{a, b\}$ of P there are $\binom{3}{2} = 3$ 2-element subsets of $\{1, 2, 3, 4, 5\} \setminus \{a, b\}$, and so P is 3-regular. Since there is no set of three pairwise disjoint 2-element subsets of $\{1, 2, 3, 4, 5\}$, P has no 3-cycles. Two non-adjacent vertices $\{a, b\}$ and $\{b, c\}$ of P have unique common neighbour $\{1, 2, 3, 4, 5\} \setminus \{a, b, c\}$. Since non-adjacent vertices in a 4-cycle have two common neighbours, P has no 4-cycles. Since P has no 3-cycles nor 4-cycles, and since $(\{1, 2\}, \{3, 4\}, \{1, 5\}, \{2, 3\}, \{4, 5\})$ (for example) is a 5-cycle, P has girth 5.

For a contradiction, suppose P has a Hamilton cycle $H = (x_1, x_2, \dots, x_{10})$. Since H is a 2-regular spanning subgraph of P , the graph $P - E(H)$ is a 1-factor F of P . Since there are no 3-cycles nor 4-cycles in P , the edges in F join vertices that are at distance 4 or 5 in H . Since there are no 4-cycles in P , the five edges of F are not $x_1x_6, x_2x_7, x_3x_8, x_4x_9$ and x_5x_{10} . Thus, there is at least one edge of F that joins vertices that are at distance 4 in H . Without loss of generality, we can assume that $x_1x_5 \in E(F)$. But then it is not possible to choose the edge of F that is incident with x_6 without forming a 3-cycle or 4-cycle. Hence there is no Hamilton cycle in P .

Now, for a contradiction, suppose P has a proper 3-edge colouring. Each colour class is a 1-factor in P , and so the union of two colour classes is a 2-factor in which each cycle has even length. Since there are no 4-cycles and no Hamilton cycles, this is impossible. Thus, P has no proper 3-edge colouring. $\square$

Chapter 2

Extremal Graphs

2.1 Turán's Theorem

Definition 2.1.1. The **Turán graph** $T_{n,r}$ is the n -vertex complete r -partite graph in which any two parts differ in cardinality by at most 1.

Theorem 2.1.2. (Turán's Theorem) The maximum number of edges in an n -vertex K_{r+1} -free simple graph is $e(T_{n,r})$, and any n -vertex K_{r+1} -free simple graph having $e(T_{n,r})$ edges is isomorphic to $T_{n,r}$.

Proof The proof is by induction on r . The theorem holds for $r = 1$ because the only n -vertex K_2 -free graph is the empty n -vertex graph. Thus, we have $r \geq 2$ for the inductive step.

Let $r \geq 2$, let G be an n -vertex K_{r+1} -free simple graph of maximum size, let v be a vertex of maximum degree $\Delta = \Delta(G)$, let G' be the graph induced by $N(v)$, and let $S = V(G) \setminus N(v)$. Note that since G is K_{r+1} -free and v is joined to every vertex of G' , the graph G' is K_r -free.

The sum $\sum_{x \in S} \deg_G(x)$ counts each edge having exactly one endpoint in S once, and counts each edge having exactly two endpoints in S twice. So

$$\sum_{x \in S} \deg_G(x) \geq e(G) - e(G'). \quad (2.1)$$

Also, since $|S| = n - \Delta$ and $\deg_G(x) \leq \Delta$ for each $x \in S$, we have

$$\sum_{x \in S} \deg_G(x) \leq \Delta(n - \Delta). \quad (2.2)$$

Thus, by (2.1) and (2.2) we have

$$\Delta(n - \Delta) + e(G') \geq e(G) \quad (2.3)$$

with equality if and only if the subgraph of G induced by S is empty and $N(x) = V(G')$ for each $x \in S$.

Now, consider the graph H obtained by adding $n - \Delta$ new vertices to $T_{\Delta, r-1}$ and joining each new vertex to each vertex of $T_{\Delta, r-1}$. The graph H is r -partite and hence is K_{r+1} -free, and

$$e(H) = \Delta(n - \Delta) + e(T_{\Delta, r-1}). \quad (2.4)$$

By the maximality of G , $e(G) \geq e(H)$ and so by (2.3) and (2.4) we have $e(G') \geq e(T_{\Delta, r-1})$. Thus, since G' is K_r -free, by the inductive hypothesis we have $G' \cong T_{\Delta, r-1}$. This implies we have equality in (2.3) and so the subgraph of G induced by S is empty and $N(x) = V(G')$ for each $x \in S$. Thus, $G \cong H$. In particular, G is a complete r -partite graph.

If $G \not\cong T_{n, r}$, then there exist parts A and B in G with $|A| \geq |B| + 2$. Let $a \in A$. If we remove the edge ab for each $b \in B$, and join a to each vertex of $A \setminus \{a\}$, then the result is a complete r -partite graph (with the same parts as G except that we have parts $A \setminus \{a\}$ and $B \cup \{a\}$ instead of parts A and B) that has more edges than G , because we removed $|B|$ edges and added $|A| - 1$. Any r -partite graph is K_{r+1} -free so our new graph contradicts the maximality of G . We conclude that $G \cong T_{n, r}$ and the proof follows by induction. $\square$

2.2 Ramsey's Theorem

Theorem 2.2.1. (Ramsey's Theorem) Let $m_1, m_2, \dots, m_k$ and r be given positive integers. There exists an integer $N = N(m_1, m_2, \dots, m_k, r)$ such that if $P = \{P_1, P_2, \dots, P_k\}$ is any partition of the r -subsets of $\{1, 2, \dots, N\}$, then there is an $i \in \{1, 2, \dots, k\}$ such that P_i contains every r -subset of some m_i -subset of $\{1, 2, \dots, N\}$.

Remark: Note that if we put $r = 1$ in Ramsey's Theorem, then we obtain the pigeonhole principle. That is, for any given positive integers $m_1, m_2, \dots, m_k$, there exists an integer N such that if $P = \{P_1, P_2, \dots, P_k\}$ is any partition of $\{1, 2, \dots, N\}$, then there is a part P_i containing at least m_i elements. In this case, $N = (m_1 - 1) + (m_2 - 1) + \dots + (m_k - 1) + 1 = m_1 + m_2 + \dots + m_k - k + 1$ suffices. So Ramsey's Theorem can be seen as a generalisation of the pigeonhole principle.

We can think of the 2-element subsets of $\{1, 2, \dots, N\}$ as the edges of the complete graph on vertices $1, 2, \dots, N$. Thus, an equivalent statement of the $r = 2$ case of Ramsey's Theorem is that for any given positive integers $m_1, m_2, \dots, m_k$, there is an integer $N = N(m_1, m_2, \dots, m_k)$ such that any edge colouring of K_N with colours $c_1, c_2, \dots, c_k$ contains an m_i -clique all of whose edges are assigned colour i .

Restricting further to the case $k = 2$, we have that for positive integers m and n , there is an integer $N = N(m, n)$ such that any red and blue edge colouring of K_N contains either a red K_m or a blue K_n . And restricting further still to the case $m = n$ we obtain that for each positive integer m there is an integer $N = N(m)$ such that any red and blue edge colouring of K_N contains a monochromatic K_m . Equivalently, we have that for each positive integer m there is an integer $N = N(m)$ such that every N -vertex graph or its complement contains an m -clique.

Definition 2.2.2. The Ramsey number $R(m_1, m_2, \dots, m_k)$ is the least positive integer N such that any edge colouring of K_N with colours $c_1, c_2, \dots, c_k$ contains an m_i -clique all of whose edges are assigned colour c_i . If $k = 2$ and $m_1 = m_2 = m$, then we denote $R(m_1, m_2)$ by $R(m)$.

It is easy to see that if $m_i = 1$ for some i , then $R(m_1, m_2, \dots, m_k) = 1$ and that if $m_i = 2$ (and $k \geq 2$), then $R(m_1, m_2, \dots, m_k) = R(m_1, m_2, \dots, m_{i-1}, m_{i+1}, \dots, m_k)$. So the Ramsey number $R(m_1, m_2, \dots, m_k)$ is interesting only if $m_1, m_2, \dots, m_k \geq 3$. The smallest interesting Ramsey number is thus $R(3)$.

Theorem 2.2.3. $R(3) = 6$.

Proof We can colour the edges of K_5 red and blue such that there is no monochromatic K_3 by colouring the edges of a 5-cycle red and the remaining edges (which also form the edges of a 5-cycle) blue. Thus, $R(3) \geq 6$.

Consider a red and blue edge colouring of K_6 . A vertex x in K_6 is incident with five edges, so without loss of generality at least three of these five edges are red. Suppose xy_1, xy_2 and xy_3 are red. Then either one of y_1y_2, y_1y_3, y_2y_3 is red and we have a red K_3 , or each of y_1y_2, y_1y_3, y_2y_3 is blue and we have a blue K_3 . Thus, in any red and blue edge colouring of K_6 there is a monochromatic K_3 and so $R(3) \leq 6$. $\square$

Theorem 2.2.4. $R(3, 4) = 9$.

Proof If we assign blue to the edges of the 8-cycle $(0, 1, \dots, 7)$ and the two 4-cycles $(0, 2, 4, 6)$ and $(1, 3, 5, 7)$, and assign red to every other edge of the complete graph on $\{0, 1, \dots, 7\}$, then there is no red K_3 and no blue K_4 . Thus, $R(3, 4) \geq 9$.

For a contradiction, suppose there exists a red and blue edge colouring of K_9 with no red K_3 and no blue K_4 . Let G_r be the graph induced by the red edges and let G_b be the graph induced by the blue edges. If G_r has a vertex x of degree at least 4, then either one of the edges of the induced graph on $N_{G_r}(x)$ is red and we have a red K_3 , or every edge of the induced graph on $N_{G_r}(x)$ is blue and we have a blue K_4 . Thus G_r has maximum degree at most 3.

Suppose G_b has a vertex x of degree at least 6. By Theorem 2.2.3 the induced graph on $N_{G_b}(x)$ has either a red K_3 or a blue K_3 . But a blue K_3 in the induced graph on $N_{G_b}(x)$ yields a blue K_4 in the induced graph on $x \cup N(x)$. Thus, G_b has maximum degree at most 5.

Since K_9 is regular of degree 8, $\deg_{G_r}(x) + \deg_{G_b}(x) = 8$ for each vertex x . Thus, since G_r has maximum degree at most 3 and G_b has maximum degree at most 5, it must be that G_r is 3-regular. But there is no 9-vertex 3-regular graph, so we have a contradiction. Thus, $R(3, 4) \leq 9$. $\square$

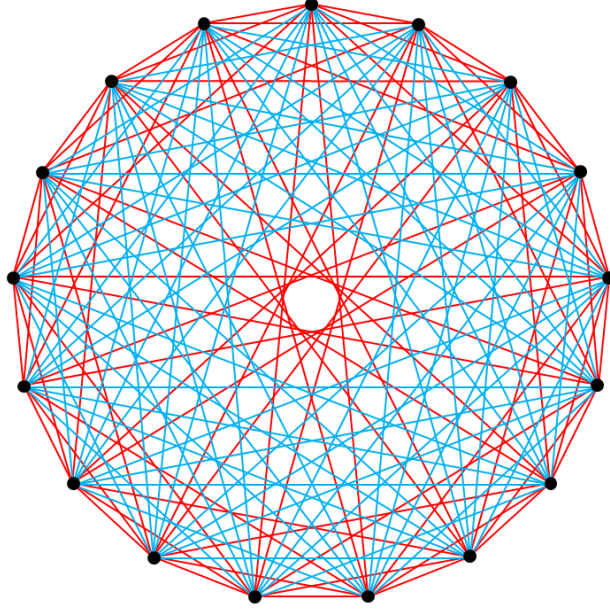
Theorem 2.2.5. $R(4) = 18$.

Proof If we assign red to the edges of the 8-cycles

$$(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16) \quad (0, 2, 4, 6, 8, 10, 12, 14, 16, 1, 3, 5, 7, 9, 11, 13, 15)$$

$$(0, 4, 8, 12, 16, 3, 7, 11, 15, 2, 6, 10, 14, 1, 5, 9, 13) \quad (0, 8, 16, 7, 15, 6, 14, 5, 13, 4, 12, 3, 11, 2, 10, 1, 9)$$

and blue to the remaining edges of the complete graph on $\{0, 1, \dots, 16\}$, then there is no monochromatic K_4 . Thus, $R(4) \geq 18$.



At each vertex of K_{18} , the number of red edges plus the number of blue edges is 17. So without loss of generality, there is a vertex x that is incident with at least 9 red edges. Let these edges be $xy_1, xy_2, \dots, xy_9$. By Theorem 2.2.4, the induced graph on $\{y_1, y_2, \dots, y_9\}$ has either a red K_3 or a blue K_4 . But a red K_3 in the induced graph on $\{y_1, y_2, \dots, y_9\}$ yields a red K_4 in the induced graph on $\{x, y_1, y_2, \dots, y_9\}$. Thus, $R(4) \leq 18$. $\square$

The arguments used in Theorems 2.2.3-2.2.5 generalise to give us the following result, an upper bound on $R(m, n)$, and thus a proof of its existence.

Theorem 2.2.6. Let $m \geq 2$ and $n \geq 2$ be integers. The Ramsey number $R(m, n) \leq R(m-1, n) + R(m, n-1)$ and if $R(m-1, n)$ and $R(m, n-1)$ are both even, then $R(m, n) \leq R(m-1, n) + R(m, n-1) - 1$. Thus, $R(m, n) \leq \binom{m+n-2}{m-1}$.

Proof Let N be a positive integer, consider an arbitrary red and blue edge colouring of K_N , let G_r be the red subgraph of K_N and let G_b be the blue subgraph of K_N .

For each vertex x , if $\deg_{G_r}(x) \geq R(m-1, n)$, then the graph induced by $N_{G_r}(x)$ contains either a red K_{m-1} or a blue K_n . If $N_{G_r}(x)$ contains a red K_{m-1} , then we have a red K_m on $N_{G_r}(x) \cup \{x\}$. Similarly, if $\deg_{G_b}(x) \geq R(m, n-1)$, then we have a red K_m or a blue K_n . Thus, to avoid a red K_m and also avoid a blue K_n we require

$$\deg_{G_r}(x) \leq R(m-1, n) - 1 \text{ and } \deg_{G_b}(x) \leq R(m, n-1) - 1 \quad (2.5)$$

for every vertex x .

Suppose $N = R(m-1, n) + R(m, n-1)$. Then $\deg_{G_r}(x) + \deg_{G_b}(x) = R(m-1, n) + R(m, n-1) - 1$ so satisfying (2.5) is impossible. Thus, $R(m, n) \leq R(m-1, n) + R(m, n-1)$.

Now suppose $R(m-1, n)$ and $R(m, n-1)$ are both even and $N = R(m-1, n) + R(m, n-1) - 1$. Then $\deg_{G_r}(x) + \deg_{G_b}(x) = R(m-1, n) + R(m, n-1) - 2$ and (2.5) is satisfied only if we have equality in both expressions. That is, $\deg_{G_r}(x) = R(m-1, n) - 1$ and $\deg_{G_b}(x) = R(m, n-1) - 1$. But since $R(m-1, n)$ and $R(m, n-1)$ are both even, we have N , $R(m-1, n) - 1$ and $R(m, n-1) - 1$ all odd. Since there is no regular graph of odd degree on an odd number of vertices, this is impossible. Thus, $R(m, n) \leq R(m-1, n) + R(m, n-1) - 1$ when $R(m-1, n)$ and $R(m, n-1)$ are both even.

The proof that $R(m, n) \leq \binom{m+n-2}{m-1}$ is by induction on $m+n$. If $m = 2$, then $R(m, n) = n$ and $\binom{m+n-2}{m-1} = \binom{n}{1} = n$, and if $n = 2$, then $R(m, n) = m$ and $\binom{m+n-2}{m-1} = \binom{m}{m-1} = m$. Thus, we can assume $3 \leq m$ and $3 \leq n$ for the inductive step. By the first part of the theorem we have $R(m, n) \leq R(m-1, n) + R(m, n-1)$ and by the inductive hypothesis we have $R(m-1, n) \leq \binom{m+n-3}{m-2}$ and $R(m, n-1) \leq \binom{m+n-3}{m-1}$. Thus we have

$$R(m, n) \leq \binom{m+n-3}{m-2} + \binom{m+n-3}{m-1} = \binom{m+n-2}{m-1}$$

and the result holds by induction. $\square$

We now give a lower bound on $R(m)$. This proof serves to demonstrate an important technique known as the *probabilistic method*.

Theorem 2.2.7. $R(m) > 2^{\frac{m}{2}-1}$

Proof The result holds for $m = 1$ and $m = 2$ so we can assume $m \geq 3$. Let $N \leq 2^{\frac{m}{2}-1}$ and colour each edge of K_N independently with probability $\frac{1}{2}$ of being red and probability $\frac{1}{2}$ of being blue. Then let X be the random variable equal to the number of monochromatic K_m in our K_N . Equivalently, X is the number of monochromatic K_m in a red and blue edge colouring of K_N that is chosen at random from the set $\mathcal{C}$ of all possible $2^{\binom{N}{2}}$ red and blue edge colourings of K_N .

Thus, the expected value of X is given by

$$\text{Exp}(X) = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} f(c)$$

where $f(c)$ denotes the number of monochromatic K_m in the colouring c . Observe that since each $f(c)$ is a nonnegative integer, if $\text{Exp}(X) < 1$, then there exists at least one c such that $f(c) = 0$. Thus, we aim to show that $\text{Exp}(X) < 1$.

For any given m -set S of vertices of K_N , the probability that S induces a red K_m is $2^{-\binom{m}{2}}$, the probability that S induces a blue K_m is also $2^{-\binom{m}{2}}$, and so the probability that S induces a monochromatic K_m is $2 \cdot 2^{-\binom{m}{2}}$. Since there are $\binom{N}{m}$ choices for S , the expected number of monochromatic K_m is given by

$$\text{Exp}(X) = \binom{N}{m} \cdot 2 \cdot 2^{-\binom{m}{2}}.$$

Since $N \leq 2^{\frac{m}{2}-1}$ and $\binom{N}{m} < N^m$ (this second inequality holds because the number of m -subsets of an N -set is less than the number of ordered m -tuples of elements of an N -set), we have

$$\binom{N}{m} < N^m \leq (2^{\frac{m}{2}-1})^m = 2^{\frac{m^2}{2}-m} = 2^{\frac{m^2}{2}-\frac{m}{2}-\frac{m}{2}} = 2^{\binom{m}{2}} \cdot 2^{-\frac{m}{2}}.$$

Thus,

$$\text{Exp}(X) < 2^{\binom{m}{2}} \cdot 2^{-\frac{m}{2}} \cdot 2 \cdot 2^{-\binom{m}{2}} = 2^{1-\frac{m}{2}} < 1$$

(since $m \geq 3$). As noted above, $\text{Exp}(X) < 1$ implies that there is a red and blue edge colouring of K_N with no monochromatic K_m . We have shown that for $N \leq 2^{\frac{m}{2}-1}$ there is a red and blue edge colouring of K_N with no monochromatic K_m . Thus, $R(m) > 2^{\frac{m}{2}-1}$. $\square$

2.3 Moore Graphs

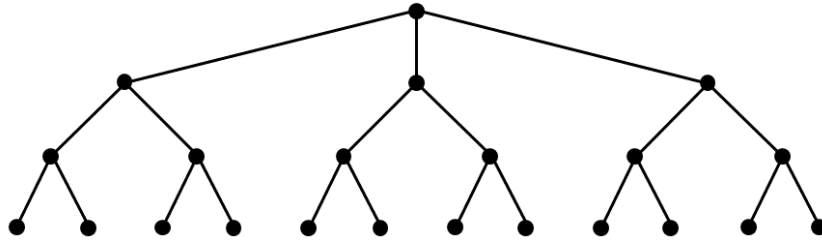
Definition 2.3.1. Let G be a graph, let v be a vertex in G and let $d \geq 0$ be an integer. We define the subgraph $G_d(v)$ of G to be the union of all paths from v having length at most d , and $S_d(v)$ to be the vertex set of $G_d(v)$. So $S_d(v) = \{x \in V(G) : d(v, x) \leq d\}$.

Definition 2.3.2. For integers $r \geq 1$ and $d \geq 0$, define the integer $M(r, d)$ by

$$M(r, 0) = 1;$$

$$M(r, 1) = 1 + r;$$

$$M(r, d) = 1 + r + r(r-1) + r(r-1)^2 + \cdots + r(r-1)^{d-1} \quad \text{for } d \geq 2.$$



Lemma 2.3.3. Let $r \geq 1$ and $d \geq 0$, let G be an r -regular graph, and let v be a vertex in G . Then

- (a) $|S_d(v)| \leq M(r, d)$; and
- (b) $|S_d(v)| = M(r, d)$ if and only if $G_d(v)$ is a tree.

Proof Simple counting shows that the number of edges in $G_d(v)$ is at most

$$r + r(r-1) + r(r-1)^2 + \cdots + r(r-1)^{d-1}$$

and we know that $G_d(v)$ is connected. It follows that $G_d(v)$ has at most $M(r, d)$ vertices with equality if and only if $G_d(v)$ is a tree. $\square$

Theorem 2.3.4. If G is an r -regular graph of diameter d , then G has at most $M(r, d)$ vertices.

Proof Let G be an r -regular graph of diameter d and let v be a vertex in G . Then every vertex of G is at distance d or less from v so the result holds by Lemma 2.3.3. $\square$

Definition 2.3.5. The upper bound from Theorem 2.3.4 on the number of vertices in a regular graph of given diameter is called the **Moore bound**. A **Moore graph** is an r -regular graph having diameter d and order $M(r, d)$.

The next theorem gives four equivalent definitions for a Moore graph.

Theorem 2.3.6. Let G be an r -regular graph with $r \geq 2$. The following are equivalent.

- (a) G has diameter d and has $M(r, d)$ vertices.
- (b) G has girth $2d + 1$ and has $M(r, d)$ vertices.
- (c) G has diameter d and girth $2d + 1$.
- (d) G has $M(r, d)$ vertices and $G_d(v)$ is a tree for each $v \in V(G)$.

Proof We show that (d) implies (a), (a) implies (b), (b) implies (c), and (c) implies (d). To do this we repeatedly use the result from Lemma 2.3.3 that $n(G_d(v)) = M(r, d)$ if and only if $G_d(v)$ is a tree.

First we show that (d) implies (a). Since $G_d(v)$ is a tree we know by Lemma 2.3.3 that $G_d(v)$ has $M(r, d)$ vertices and hence it contains all the vertices of G . It follows that G has diameter d and (a) holds.

We now show that (a) implies (b). Suppose (a) holds, let g be the girth of G , and let v be a vertex in a cycle C of length g . Diameter d implies $G_d(v)$ contains all the vertices of G . Thus, $|S_d(v)| = M(r, d)$ so $G_d(v)$ is a tree by Lemma 2.3.3. But if $g \leq 2d$, then C is a subgraph of $G_d(v)$. Thus, $g \geq 2d + 1$. Also, since $G_d(v)$ is a tree containing all the vertices of G , there exist adjacent vertices at distance d from v . This yields a cycle of length $2d + 1$, so $g = 2d + 1$ and (b) holds.

We now show that (b) implies (c). Suppose (b) holds and let u, v be vertices of G . Girth $2d + 1$ implies $G_d(v)$ is a tree. Thus, $|S_d(v)| = M(r, d)$ and $G_d(v)$ contains all the vertices of G . In particular, $d(u, v) \leq d$ and so G has diameter at most d . Girth $2d + 1$ also implies that the diameter of G is at least d . Thus, G has diameter exactly d and (c) holds.

We now show that (c) implies (d). Suppose (c) holds and let v be a vertex of G . Diameter d implies all the vertices of G are in $G_d(v)$. Also, as previously, girth $2d + 1$ implies $G_d(v)$ is a tree and hence by Lemma 2.3.3 we have $|S_d(v)| = M(r, d)$. So G has $M(r, d)$ vertices and (d) holds. $\square$

Complete graphs of order at least 3 and cycles of odd length are Moore graphs. For $r = 2$, we have $M(2, d) = 2d + 1$ and the $(2d + 1)$ -cycle, being 2-regular and having diameter d , is the unique Moore graph of degree 2 and diameter d . The Moore graphs of diameter 1 are precisely the complete graphs; K_{r+1} is the only r -regular graph of diameter 1, and it is a Moore graph because $M(r, 1) = r + 1$. This covers $d = 1$ and $r = 2$. The next smallest values of r and d to consider are $d = 2$ and $r = 3$.

Theorem 2.3.7. The Petersen graph is the unique 3-regular Moore graph of diameter 2.

Proof By Theorem 1.0.6, the Petersen Graph is a 3-regular graph of diameter $d = 2$ and girth $g = 5$. Thus, it is a Moore graph. It remains to show that it is unique.

Let G be an arbitrary 3-regular Moore graph of diameter 2. So G has $M(3, 2) = 10$ vertices and has girth 5. Let the vertices of G be $v_1, v_2, \dots, v_{10}$. Without loss of generality we can assume the neighbours of v_1 are v_2, v_3 and v_4 . The vertices v_2, v_3, v_4 are pairwise non-adjacent (otherwise we have a 3-cycle), and each has two neighbours other than v_1 . These six vertices are distinct (otherwise we have a 4-cycle). Let v_5 and v_6 be the other two neighbours of v_2 , let v_7 and v_8 be the other two neighbours of v_3 , and let v_9 and v_{10} be the other two neighbours of v_4 .

Since G has girth 5, v_5 has one neighbour from $\{v_7, v_8\}$ and one neighbour from $\{v_9, v_{10}\}$. Without loss of generality we can assume that v_5v_7 and v_5v_9 are edges of G . The vertex v_6 has two neighbours other than v_2 , and it follows (again from G having girth 5) that v_6v_8 and v_6v_{10} are edges of G . Girth 5 forces the remaining two edges to be v_7v_{10} and v_8v_9 .

The mapping

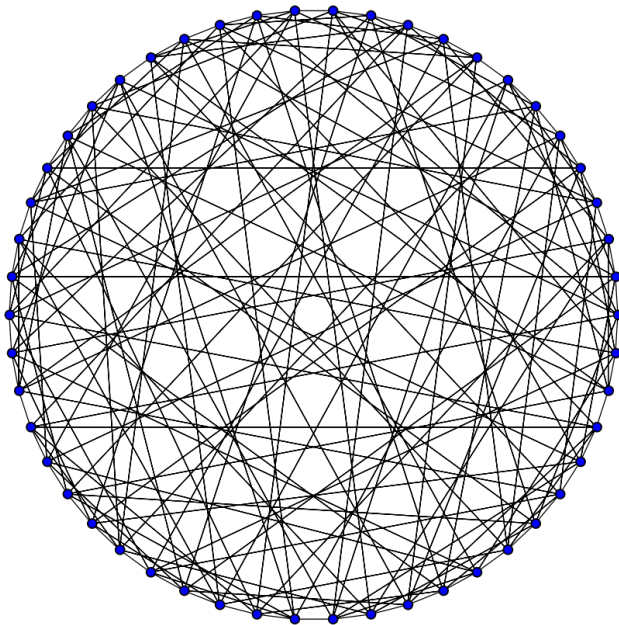
$$\begin{aligned} v_1 &\mapsto \{1, 2\} & v_2 &\mapsto \{3, 4\} & v_3 &\mapsto \{3, 5\} & v_4 &\mapsto \{4, 5\} & v_5 &\mapsto \{1, 5\} \\ v_6 &\mapsto \{2, 5\} & v_7 &\mapsto \{2, 4\} & v_8 &\mapsto \{1, 4\} & v_9 &\mapsto \{2, 3\} & v_{10} &\mapsto \{1, 3\} \end{aligned}$$

is an isomorphism from G to the Petersen graph (as given in Definition 1.0.5). □

Another Moore graph is the *Hoffman-Singleton graph* which was constructed in 1960 [9].

Definition 2.3.8. The **Hoffman-Singleton graph** has vertex set $\mathbb{Z}_2 \times \mathbb{Z}_5 \times \mathbb{Z}_5$ and has

- $(0, i, j)$ joined to $(0, i, j + 1)$ for each $i, j \in \mathbb{Z}_5$;
- $(1, i, j)$ joined to $(1, i, j + 2)$ for each $i, j \in \mathbb{Z}_5$;
- $(0, i, j)$ joined to $(1, k, \ell)$ if and only if $j = ik + \ell$.



The Hoffman-Singleton graph.

Theorem 2.3.9. The Hoffman-Singleton graph is a 50-vertex 7-regular Moore graph of diameter 2 and girth 5.

Proof Omitted. □

The following theorem combines results from [2], [4] and [9]. In [9] it is shown that for degree at least 3 and for diameters 2 and 3, the only Moore graphs are the Petersen graph and the Hoffman-Singleton graph, with the possible exception that there may exist a 3250-vertex 57-regular Moore graph of diameter 2 and girth 5. In [2] and [4] it is shown that there are no Moore graphs of degree at least 3 and diameter at least 4.

Theorem 2.3.10. The Moore graphs are

- The complete graph of order $r + 1$ is an r -regular Moore graph of diameter 1 and girth 3.
- The cycle of length $2d + 1$ is a 2-regular Moore graph of diameter d and girth $2d + 1$.
- The Petersen graph is a 10-vertex 3-regular Moore graph of diameter 2 and girth 5.
- The Hoffman-Singleton graph is a 50-vertex 7-regular Moore graph of diameter 2 and girth 5.
- It is unknown whether there exists a 3250-vertex 57-regular Moore graph of diameter 2 and girth 5.
- There are no more Moore graphs.

Proof Omitted. □

2.4 Cages

Definition 2.4.1. Let $r \geq 2$ and $g \geq 3$. An (r, g) -**graph** is an r -regular graph that has girth g . An (r, g) -**cage** is an (r, g) -graph of smallest order.

The following result was proved by Erdős and Sachs in 1963 [6]. A proof is in [3].

Theorem 2.4.2. For $r \geq 2$ and $g \geq 3$, there exists an (r, g) -graph, and hence an (r, g) -cage.

Proof Omitted. □

Definition 2.4.3. Define the integer $M'(r, g)$ by

$$M'(r, g) = \begin{cases} 1 + r + r(r-1) + r(r-1)^2 + \cdots + r(r-1)^{(g-3)/2} & \text{if } g \text{ is odd} \\ 2 \left(1 + (r-1) + (r-1)^2 + \cdots + (r-1)^{(g-2)/2} \right) & \text{if } g \text{ is even.} \end{cases}$$

Theorem 2.4.4. Any (r, g) -graph has at least $M'(r, g)$ vertices, and if g is odd, then an (r, g) -graph having exactly $M'(r, g)$ vertices is a Moore graph.

Proof Let G be an (r, g) -graph and let v be an arbitrary vertex in G . First suppose g is odd, let $g = 2d + 1$, and note that $M'(r, g) = M(r, d)$. Girth $2d + 1$ implies $G_d(v)$ is a tree, and so by Lemma 2.3.3 we know that the number of vertices in G is at least $M(r, d) = M'(r, g)$. Further, if G has exactly $M'(r, g)$ vertices, then we know by Theorem 2.3.6 that G is a Moore graph.

Now suppose g is even. Let $g = 2d$, let uv be an edge of G and let $G'_{d-1}(uv)$ be the union of all the paths of length $d - 1$ from u and all the paths of length $d - 1$ from v . Girth $2d$ implies that $G'_{d-1}(uv)$ is a tree. Since G is r -regular, the number of vertices in $G'_{d-1}(uv)$ is

$$2 \left(1 + (r-1) + (r-1)^2 + \cdots + (r-1)^{d-1} \right).$$

Since $d - 1 = (g - 2)/2$ this expression equals $M'(r, g)$, and so since every vertex of $G'_{d-1}(uv)$ is in G , the number of vertices in G is at least $M'(r, g)$.

To see that the number of vertices in $G'_{d-1}(uv)$ is indeed as claimed above, consider the two trees obtained from $G'_{d-1}(uv)$ by deleting the edge uv . The tree containing u has $r - 1$ vertices at distance 1 from u , $(r - 1)^2$ vertices at distance 2 from u , and so on up to $(r - 1)^{d-1}$ vertices at distance $d - 1$ from u . Thus there are $1 + (r - 1) + (r - 1)^2 + \cdots + (r - 1)^{d-1}$ in the tree containing u , and there are the same number of vertices in the tree containing v . □

Remark: When g is even, (r, g) -cages of order $M'(r, g)$ are sometimes also called Moore graphs.

For each $g \geq 3$, the cycle of length g is the only $(2, g)$ -cage. For each $r \geq 2$, the complete graph of order $r + 1$ is the only $(r, 3)$ -cage. Next we consider $(r, 4)$ -cages.

Theorem 2.4.5. For each $r \geq 2$, the complete bipartite graph $K_{r,r}$ is the unique $(r, 4)$ -cage.

Proof By Theorem 2.4.4 any $(r, 4)$ -graph has at least $M'(r, 4) = 2r$ vertices. Thus, since $K_{r,r}$ is r -regular and has girth 4, it is an $(r, 4)$ -cage. It remains to show that it is the only $(r, 4)$ -cage.

Let G be an $(r, 4)$ -cage. Then G has $2r$ vertices. Let u_1 be a vertex of G , let $v_1, v_2, \dots, v_r$ be the neighbours of u_1 and let $u_1, u_2, \dots, u_r$ be the neighbours of v_1 . Since G has no 3-cycles, $v_1, v_2, \dots, v_r$ are pairwise non-adjacent, $u_1, u_2, \dots, u_r$ are pairwise non-adjacent, and $\{u_1, \dots, u_r\} \cap \{v_1, \dots, v_r\} = \emptyset$. Thus, $\{u_1, \dots, u_r\} \cup \{v_1, \dots, v_r\}$ contains all the vertices of G , and since G is r -regular, each u_i is adjacent to every v_j . So $G \cong K_{r,r}$. $\square$

Next we consider $(3, g)$ -cages.

Remark: A $(3, g)$ -cage is sometimes called a g -cage.

We have already seen that K_4 is the unique $(3, 3)$ -cage and that $K_{3,3}$ is the unique $(3, 4)$ -cage.

Theorem 2.4.6. The Petersen graph is the unique $(3, 5)$ -cage.

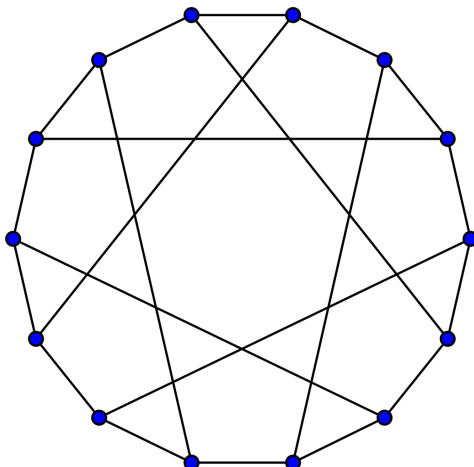
Proof The uniqueness of the Petersen graph as a 10-vertex 3-regular graph of girth 5 is proved in Theorem 2.3.7. $\square$

For $g \in \{6, 7, 8, 11, 12\}$ there is a unique $(3, g)$ -cage, and these have orders 14, 24, 30, 112 and 126 respectively. The number of nonisomorphic 9-cages is exactly 18, and these have order 58. There are exactly 3 nonisomorphic 10-cages, and these have order 70. No $(3, g)$ -cage is known for $g \geq 13$. The $(3, 6)$ -cage is the *Heawood graph*, the $(3, 7)$ -cage is the *McGee graph*, and the $(3, 8)$ -cage is the *Tutte-Coxeter graph*.

Definition 2.4.7. Let

$$\begin{aligned} S &= \{\{0 + i, 1 + i, 3 + i\} : i \in \mathbb{Z}_7\} \\ &= \{\{0, 1, 3\}, \{1, 2, 4\}, \{2, 3, 5\}, \{3, 4, 6\}, \{4, 5, 0\}, \{5, 6, 1\}, \{6, 0, 2\}\}. \end{aligned}$$

The **Heawood graph** has vertex set $\mathbb{Z}_7 \cup S$ and edge set given by joining each $\{a, b, c\} \in S$ to a , b and c .



The Heawood graph.

Theorem 2.4.8. The Heawood graph is a 14-vertex 3-regular bipartite graph with girth 6, and is the unique $(3, 6)$ -cage.

Proof It is easy to see the Heawood graph has 14 vertices, is 3-regular, and is bipartite with parts S and $\mathbb{Z}_7$. Thus, since $M'(3, 6) = 14$, if the Heawood graph has girth 6, then it is a $(3, 6)$ -cage. The proof that it is unique is omitted. The cycle

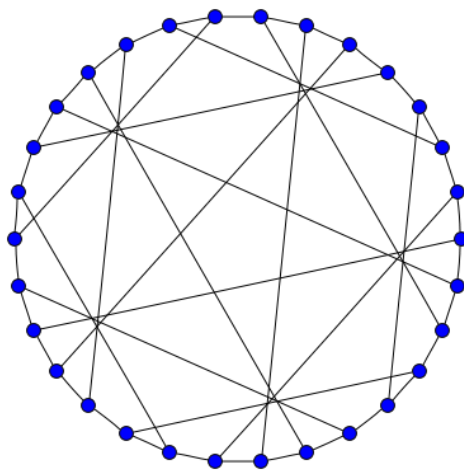
$$(0, \{0, 1, 3\}, 1, \{1, 2, 4\}, 2, \{6, 0, 2\})$$

is a 6-cycle in the Heawood graph. So it remains only to show that there is no cycle of length less than 6.

Suppose for a contradiction that the Heawood graph has a cycle of length less than 6. Since it is bipartite this implies it contains a 4-cycle. By construction, this means that we have two elements $a, b \in \mathbb{Z}_7$ and two triples $A, B \in S$ such that $a, b \in A$ and $a, b \in B$. However, it can be seen that each pair of elements of $\mathbb{Z}_7$ is contained in exactly one triple of S . $\square$

The set S in Definition 2.4.7 is a *projective plane of order 2*, also known as a *Fano plane*. If we think of the elements of $\mathbb{Z}_7$ as points and the triples of S as lines, then there is a unique line joining any two points, and any two lines meet in a unique point. This idea generalises. There exists an $(r, 6)$ -cage of order $2(1 + (r - 1) + (r - 1)^2)$ if and only if there exists a projective plane of order $r - 1$. It is known that there exists a projective plane of order q whenever q is a power of a prime, but it is unknown whether there exist any projective planes of non-prime power order. There is no projective plane of order 6 and no projective plane of order 10.

Definition 2.4.9. Let A be the set of 2-element subsets of $\{1, 2, 3, 4, 5, 6\}$ and let B be the set of all partitions of $\{1, 2, 3, 4, 5, 6\}$ into pairs. The **Tutte-Coxeter graph** is the bipartite graph with parts A and B and with $a \in A$ adjacent to $b \in B$ if and only if $a \in b$.



The Tutte-Coxeter graph.

Theorem 2.4.10. The Tutte-Coxeter graph is a 30-vertex 3-regular bipartite graph with girth 8, and is the unique $(3, 8)$ -cage.

Proof It is easy to see that the Tutte-Coxeter graph has $\binom{6}{2} + \left(\binom{6}{2}\binom{4}{2}/3!\right) = 30$ vertices and is 3-regular and bipartite. Thus, since $M'(3, 8) = 30$, if the Tutte-Coxeter graph has girth 8, then it is a $(3, 8)$ -cage. The proof that it is unique is omitted. Denote each 2-element subset $\{a, b\}$ of $\{1, 2, 3, 4, 5, 6\}$ by ab . The graph contains, for example, the 8-cycle

$$(12, \{12, 34, 56\}, 34, \{15, 26, 34\}, 15, \{15, 23, 46\}, 46, \{12, 35, 46\})$$

and so has girth at most 8. Since it is bipartite, it remains only to show that it has no 4-cycles and no 6-cycles.

A 4-cycle would necessarily be of the form $(ab, \{ab, cd, **\}, cd, \{ab, cd, **\})$ with a, b, c, d distinct, but there is only one partition of $\{1, 2, 3, 4, 5, 6\}$ into pairs that contains both ab and cd . A 6-cycle would necessarily be of the form $(ab, \{ab, cd, **\}, cd, \{cd, ef, **\}, ef, \{ab, ef, **\})$. But this implies a, b, c, d, e, f are all distinct and hence that the vertices of B in the cycle are all equal, which is impossible. $\square$

See Wolfram Mathworld [\[21\]](#) or Wikipedia [\[23\]](#) for more information on known cages.

Chapter 3

Connectivity

In this section we assume that all our graphs have no loops.

Definition 3.0.1. A **vertex cut**, or just **cut**, of a graph G is a proper subset $S \subset V(G)$ such that $G - S$ is disconnected. A cut of cardinality k is called a **k -cut**. For $x, y \in V(G)$, an **x, y -cut** is a set $S \subseteq V(G) \setminus \{x, y\}$ such that $G - S$ has no x, y -path. The **connectivity** of G is the minimum cardinality of a set S of vertices such that $G - S$ is disconnected or has only one vertex. The connectivity of G is denoted by $\kappa(G)$. A graph is **k -connected** if $\kappa(G) \geq k$.

For a complete graph G , if S is any proper subset of $V(G)$, then $G - S$ is connected (complete graphs have no cuts) and so $\kappa(G) = n(G) - 1$. For any graph G that is not a complete graph, there exist distinct non-adjacent vertices x and y . So $V(G) \setminus \{x, y\}$ is either empty or is an x, y -cut of G . Thus, for a non-complete graph G with at least 2 vertices we have $\kappa(G) \leq n(G) - 2$ and we have a vertex cut of cardinality $\kappa(G)$.

Definition 3.0.2. Let G be a graph and let S and T be subsets of $V(G)$. The complement of S is denoted by $\overline{S}$ and defined by $\overline{S} = V(G) \setminus S$. The set of edges having one endpoint in S and the other in T is denoted by $[S, T]$.

Definition 3.0.3. An **edge cut** is a set of edges of the form $[S, \overline{S}]$ where S is a non-empty proper subset of $V(G)$. For $x, y \in V(G)$, an **x, y -edge cut** is an edge cut $[S, T]$ in G such that $x \in S$ and $y \in T$ (which implies that there is no x, y -path in $G - [S, T]$). For a graph G with at least two vertices, the **edge connectivity** of G is the minimum cardinality of an edge cut. An edge cut of cardinality k is called a **k -edge cut**. The edge connectivity of G is denoted by $\kappa'(G)$. A graph is **k -edge-connected** if $\kappa'(G) \geq k$.

Theorem 3.0.4. Let G be a graph. Then

$$\kappa(G) \leq \kappa'(G) \leq \delta(G).$$

Proof The edges incident with a vertex of minimum degree form an edge cut so $\kappa'(G) \leq \delta(G)$ and it remains to show that $\kappa(G) \leq \kappa'(G)$.

Consider an edge cut $[S, \bar{S}]$ of cardinality $\kappa'(G)$. If every vertex of S is adjacent to every vertex of $\bar{S}$, then

$$\kappa'(G) = |[S, \bar{S}]| \geq |S||\bar{S}| \geq n(G) - 1 \geq \kappa(G).$$

We noted just after giving the definition of $\kappa(G)$ (Definition 3.0.1) that $\kappa(G) \leq n(G) - 1$.

Otherwise there exist non-adjacent vertices $x \in S$ and $y \in \bar{S}$. Let T consist of all the neighbours of x in $\bar{S}$ and all the vertices of $S \setminus \{x\}$ that have neighbours in $\bar{S}$. Then every x, y -path contains a vertex of T . So T is a cut and $\kappa(G) \leq |T|$.

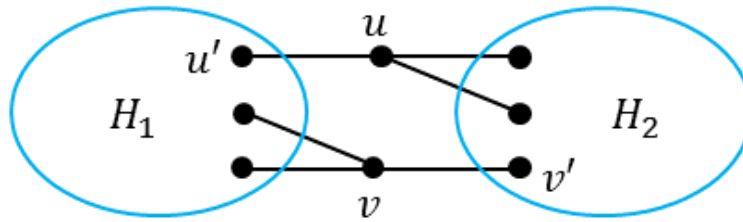
Now, for each vertex v in $T \cap \bar{S}$ there is an edge of $[S, \bar{S}]$ joining v to x , and for each vertex u in $T \cap S$, there is an edge of $[S, \bar{S}]$ incident with u . The $|T|$ edges just mentioned are pairwise distinct so $|T| \leq |[S, \bar{S}]|$. Thus, we have $\kappa(G) \leq |T| \leq |[S, \bar{S}]| = \kappa'(G)$. $\square$

Theorem 3.0.5. Let G be a 3-regular graph with more than two vertices. Then $\kappa(G) = \kappa'(G)$.

Proof By Theorem 3.0.4, we have $\kappa(G) \leq 3$. If $\kappa(G) = 0$, then G is not connected and $\kappa(G) = \kappa'(G) = 0$. If $\kappa(G) = 3$, then by Theorem 3.0.4, we have $\kappa(G) = \kappa'(G) = 3$. This leaves the cases $\kappa(G) = 1$ and $\kappa(G) = 2$. Note that we thus have a cut of cardinality $\kappa(G)$ because $\kappa(G) \in \{1, 2\}$ and G has more than two vertices together imply that G has a pair of non-adjacent vertices.

Suppose $\kappa(G) = 1$ and let $\{v\}$ be a cut. Then some component of $G - v$ has exactly one neighbour v' of v , and so vv' is an edge cut of G . Thus, $\kappa'(G) = 1 = \kappa(G)$.

Now suppose $\kappa(G) = 2$ and let $\{u, v\}$ be a cut. Note that $\kappa'(G) \geq 2$ by Theorem 3.0.4. For a contradiction, suppose that there is no edge cut of cardinality 2. Thus, each component of $G - \{u, v\}$ is joined to $\{u, v\}$ by at least 3 edges. It follows that $G - \{u, v\}$ has exactly 2 components H_1 and H_2 , and (without loss of generality) that u has 1 neighbour in H_1 and 2 neighbours in H_2 and v has 2 neighbours in H_1 and 1 neighbour in H_2 . Thus, u is not adjacent to v and $\{uu', vv'\}$ is an edge cut of G where u' is the neighbour of u in H_1 and v' is the neighbour of v in H_2 . $\square$



3.1 Network Flows

Definition 3.1.1. A **network** consists of a digraph G , a function $c : E(G) \rightarrow \mathbb{R}$ such that $c(e) \geq 0$ for each $e \in E(G)$, and two distinguished vertices s and t . The vertex s is the **source**, the vertex t is the **sink**, and $c(e)$ is the **capacity** of the edge e .

Definition 3.1.2. Let G be a network. A **flow** is any function $f : e(G) \rightarrow \mathbb{R}$. A flow is **feasible** if it satisfies

$$0 \leq f(e) \leq c(e) \text{ for each } e \in E(G) \quad (3.1)$$

and

$$f^+(v) = f^-(v) \text{ for each } v \in V(G) \setminus \{s, t\}, \quad (3.2)$$

where $f^+(v) = \sum_{x \in N^+(v)} f(vx)$ is the total flow out of v , and $f^-(v) = \sum_{x \in N^-(v)} f(xv)$ is the total flow into v . The conditions in (3.1) are called the **capacity constraints** and the conditions in (3.2) are called the **conservation constraints**. The **value** of a flow f , denoted by $\text{val}(f)$, is given by

$$\text{val}(f) = f^-(t) - f^+(t).$$

In general, $f^+(v) - f^-(v)$ is called the **net flow out of v** , and $f^-(v) - f^+(v)$ is called the **net flow into v** . So the value of a flow is the net flow into the sink. The conservation constraints say that for each vertex v other than the source and the sink, the net flow out of v is zero.

For any set U of vertices in a network with flow f , the **total flow out of U** is the sum of the flows on the edges leaving U and is denoted by $f^+(U)$, the **total flow into U** is the sum of the flows on the edges entering U and is denoted by $f^-(U)$, the **net flow out of U** is $f^+(U) - f^-(U)$, and the **net flow into U** is $f^-(U) - f^+(U)$.

Lemma 3.1.3. If U is a set of vertices in a network with flow f , then the net flow out of U is the sum of the net flows out of the vertices of U , and the net flow into U is the sum of the net flows into the vertices of U . That is,

$$f^+(U) - f^-(U) = \sum_{u \in U} (f^+(u) - f^-(u)) \quad (3.3)$$

and

$$f^-(U) - f^+(U) = \sum_{u \in U} (f^-(u) - f^+(u)). \quad (3.4)$$

Proof Consider the contribution of the flow $f(e)$ on an edge $e = xy$ to each side of (3.3). If $x, y \in U$, then the contribution is zero on the left, and the contribution on the right is plus $f(e)$ at x and $-f(e)$ at y . If $x, y \notin U$, then there is no contribution to either side. If $x \in U$ and $y \notin U$, then the contribution on the left is plus $f(e)$ and the contribution on the right is plus $f(e)$ at x . If $x \notin U$ and $y \in U$, then the contribution on the left is minus $f(e)$ and the contribution on the right is minus $f(e)$ at y . In all cases the contribution to each side is the same. Equation (3.4) follows by multiplying each side of (3.3) by -1 . $\square$

An important problem is to determine a feasible flow of maximum value in a network. Such a flow is called a **maximum feasible flow**, or just **maximum flow**.

Definition 3.1.4. Let f be a feasible flow in a network N , let P be an s, t -path in the underlying graph of N , and let

- $\epsilon(e) = c(e) - f(e)$ if e is traversed in the forward direction when travelling from s to t along P ;
and
- $\epsilon(e) = f(e)$ if e is traversed in the backward direction when travelling from s to t along P .

If $\epsilon(e) > 0$ for each $e \in P$, then P is an **f -augmenting path**. The **tolerance** of P is $\min_{e \in E(P)} \epsilon(e)$.

Lemma 3.1.5. Let f be a feasible flow in a network. If P is an f -augmenting path with tolerance ε , then a feasible flow f' with $\text{val}(f') = \text{val}(f) + \varepsilon$ can be obtained by

- increasing the flow by ε on e for each $e \in E(P)$ that is traversed in the forward direction when travelling from s to t along P ; and
- decreasing the flow by ε on e for each $e \in E(P)$ that is traversed in the backward direction when travelling from s to t along P .

Proof The definition of tolerance ensures $0 \leq f'(e) \leq c(e)$ on each edge. The flow is unchanged at each vertex not in P , and at each internal vertex v of P the change to the flow into v equals the change to the flow out of v . So the net flow is unchanged and remains at zero. The net flow into the sink increases by the tolerance ε (and the net flow out of the source also increases by ε). $\square$

We now examine bounds on the value of a feasible network flow.

Definition 3.1.6. In a network, a **source/sink cut**, or just **cut**, is given by a partition of the vertex set into two parts S and T with $s \in S$ and $t \in T$, and consists of all the edges from S to T . The **capacity of a cut** is sum of the capacities of the edges in the cut, and the capacity of the cut $[S, T]$ is denoted by $\text{cap}(S, T)$.

Lemma 3.1.7. If f is a feasible flow and $[S, T]$ is a cut, then each of the net flow out of S and the net flow into T equals $\text{val}(f)$. That is,

$$f^+(S) - f^-(S) = f^-(T) - f^+(T) = \text{val}(f).$$

Proof Since $\{S, T\}$ is a partition of the vertex set, the edges leaving S are precisely the edges entering T , and the edges entering S are precisely the edges leaving T . Thus, the net flow out of S equals the net flow into T . By Lemma 3.1.3, the net flow out of S is the sum of the net flows out of the vertices in S , and the conservation constraints tell us that the net flow out of each vertex in $S \setminus \{s\}$ is zero. So the net flow out of S equals the net flow out of s . Similarly, the net flow into T equals the net flow into t , which by definition is $\text{val}(f)$. $\square$

Lemma 3.1.8. The value of any feasible flow does not exceed the capacity of any cut. That is,

$$\text{val}(f) \leq \text{cap}(S, T)$$

where f is any feasible flow and $[S, T]$ is any cut.

Proof The capacity constraints tell us that the total flow into T is at most $\text{cap}(S, T)$, so the net flow into T is also at most $\text{cap}(S, T)$. But by Lemma 3.1.7, the net flow into T equals $\text{val}(f)$. Thus, $\text{val}(f) \leq \text{cap}(S, T)$. $\square$

For any given feasible flow f , the following algorithm produces either an augmenting path or a cut with capacity equal to $\text{val}(f)$ (this will be shown in the proof of Theorem 3.1.10). By Lemma 3.1.8, if

there is a cut $[S, T]$ with capacity equal to $\text{val}(f)$, then $[S, T]$ is a minimum cut (there is no cut with smaller capacity) and f is a maximum flow (there is no feasible flow of larger value). Otherwise, the augmenting path can be used to construct a feasible flow of larger value, and the algorithm can be applied again.

Algorithm 3.1.9. (Ford-Fulkerson Algorithm)

The general approach involves finding all the vertices that can be reached from the source by a path of positive tolerance (the definition of tolerance extends in an obvious way to any path starting from s). If t is reached, then an augmenting path is found. Otherwise the set S of vertices that can be reached and its complement $\bar{S}$ form a cut $[S, \bar{S}]$ of capacity $\text{val}(f)$.

Input: A network with a feasible flow f .

Output: An augmenting path for f or a cut with capacity $\text{val}(f)$.

Initialization: $R = \{s\}$ is the set of “reached” vertices. $S = \emptyset$ is the set of “searched” vertices.

Iteration:

- Choose $v \in R \setminus S$.
- For each edge $e = vx$ such that $x \notin R$ and $f(e) < c(e)$, add x to R and record that x is reached from v .
- For each edge $e = xv$ such that $x \notin R$ and $f(e) > 0$, add x to R and record that x is reached from v .
- After checking each edge incident with v , add v to the set S of searched vertices.
- If the sink t has been reached, then output the corresponding f -augmenting path and terminate.
- If $R = S$, then output the cut $[S, \bar{S}]$ and terminate.
- Otherwise, iterate.

Theorem 3.1.10. (Max-flow Min-cut Theorem) In any network, the maximum value of a feasible flow equals the minimum capacity of a source/sink cut.

Proof The flow where every edge has flow zero can be used as the initial feasible flow. Applying Algorithm 3.1.9 we either obtain $t \in R$ or $S = R$. If $t \in R$, then we have an augmenting path and can construct a new feasible flow of increased value. The process can then be repeated. If the capacities are rational, then each increase in the flow is a multiple of $1/a$ where a is the least common multiple of the denominators of the capacities. Thus, after finitely many augmentations the capacity of some cut will be reached and the algorithm will terminate with $S = R$.

When the algorithm terminates with $S = R$, we have $s \in S$ and $t \notin S$. So $[S, \bar{S}]$ is a cut. Let f be the flow. Then for each edge $e = vx$ with $v \in S$ and $x \in \bar{S}$ we have $f(e) = c(e)$ and for each edge $e = xv$ with $v \in S$ and $x \in \bar{S}$ we have $f(e) = 0$. Thus, $f^+(S) = \text{cap}(S, \bar{S})$ and $f^-(S) = 0$. But we know from Lemma 3.1.7 that $\text{val}(f) = f^+(S) - f^-(S)$ so we have $\text{val}(f) = \text{cap}(S, \bar{S})$. By Lemma

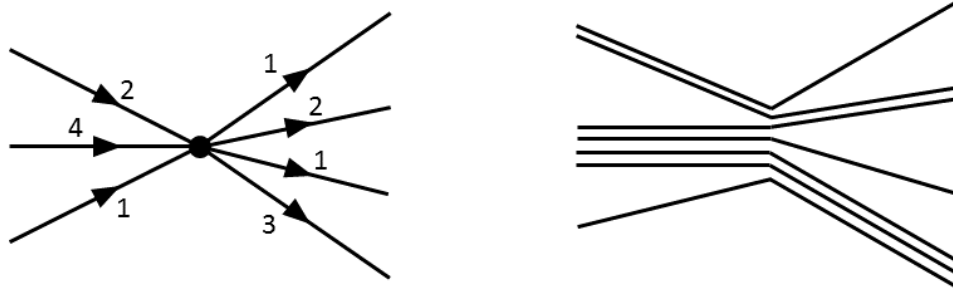
3.1.8, the value of any flow is at most $\text{cap}(S, \bar{S})$ and the capacity of any cut is at least $\text{val}(f)$. Thus f is a maximum feasible flow and $[S, \bar{S}]$ is a minimum cut.

The theorem is thus proved for rational capacities. Edmonds and Karp [5] proved that the theorem also holds for any real capacities. $\square$

Theorem 3.1.11. If all the capacities in a network are integers, then there exists a maximum feasible flow f that assigns integer flows to each edge, and there exist $\text{val}(f)$ paths from s to t such that each edge e occurs in at most $f(e)$ paths.

Proof In Algorithm 3.1.9, the change in flow value is always the flow on an edge or the difference between a capacity and a flow on an edge. Thus, starting with the zero flow, we only ever obtain integer flows and so the algorithm produces a maximum feasible flow that assigns integer flows to each edge.

Since the net flow out of any vertex other than the source and the sink is zero, the flows entering the vertex can be partitioned into flows of unit value and paired with flows of unit value leaving the vertex. If any cycles arise, then the flow on each of their edges can be reduced from one to zero without changing the flow value. Thus, we obtain $\text{val}(f)$ paths from s to t such that each edge e occurs in at most $f(e)$ paths. $\square$



3.2 Menger's Theorem

Menger [12] proved the following theorem in 1927.

Theorem 3.2.1. (Menger's Theorem) If x and y are distinct non-adjacent vertices in a graph, then the maximum number of pairwise internally disjoint x, y -paths equals the minimum cardinality of an x, y -cut.

Proof Let x and y be distinct non-adjacent vertices in a graph G , and let κ be the cardinality of a minimum x, y -cut. The theorem clearly holds if $\kappa = 0$, so we may assume $\kappa \geq 1$. Any x, y -cut must contain an internal vertex from each x, y -path. Thus, κ is greater than or equal to the maximum number of pairwise internally disjoint x, y -paths.

We need to show that there exist κ pairwise internally disjoint x, y -paths. For this we use the max-flow min-cut theorem. Construct a network N from G as follows. For each vertex v in G , in N

there are two vertices v_{in} and v_{out} and a (directed) edge $v_{in}v_{out}$ of capacity 1. For each (undirected) edge uv in G , in N there is an edge $u_{out}v_{in}$ of capacity ∞ and an edge $v_{out}u_{in}$ of capacity ∞ , where ∞ is a very large integer (any integer larger than the number of vertices in the graph, for example, will suffice). The source is x_{out} and the sink is y_{in} .

We show that the minimum capacity of a cut in N is κ . Let A be an x, y -cut of cardinality κ in G , let X be the component of $G \setminus A$ containing x , let Y be the component of $G \setminus A$ containing y , let $S = \{v_{in} : v \in V(X)\} \cup \{v_{out} : v \in V(X)\} \cup \{v_{in} : v \in A\}$, and let $T = \{v_{in} : v \in V(Y)\} \cup \{v_{out} : v \in V(Y)\} \cup \{v_{out} : v \in A\}$. It can be seen that $[S, T]$ is a cut of capacity κ .

For a contradiction suppose $[S', T']$ is a cut of capacity less than κ . It is clear that each edge of $[S', T']$ is of the form $v_{in}v_{out}$ (all other edges have capacity ∞). Consider the set $B = \{v \in V(G) : v_{in}v_{out} \in [S', T']\}$, and note that the capacity of $[S', T']$ is $|B|$. Since $|B| < \kappa$, B is not an x, y -cut in G . Let $x, z^1, z^2, \dots, z^r, y$ be an x, y -path in $G \setminus B$ ($r \geq 1$ because x and y are not adjacent in G). It follows that at least one edge e of the path

$$x_{out}, z_{in}^1, z_{out}^1, z_{in}^2, z_{out}^2, \dots, z_{in}^r, z_{out}^r, y_{in}$$

is in $[S', T']$. This is a contradiction (none of the edges of this path are in $[S', T']$ because the edges of the path with finite capacity are of the form $z_{in}^i z_{out}^i$, and each z^i is not in B), and we conclude that the minimum capacity of a cut in N is κ .

Thus, by the max-flow min-cut theorem, N has a flow of value κ , and so by Theorem 3.1.11 has a flow of value κ in which the flow on each edge is an integer. It follows from the construction of N that for each $v \in V(G) \setminus \{x, y\}$, the flow into v_{in} and the flow out of v_{out} are either both 0 or both 1. Thus, the edges uv in G such that the flow on the edge $u_{out}v_{in}$ of N is 1 form the required set of κ pairwise internally disjoint x, y -paths in G . $\square$

Theorem 3.2.2. If x and y are distinct vertices in a simple graph or digraph, then the maximum number of pairwise edge-disjoint x, y -paths equals the minimum cardinality of an x, y -edge cut.

Proof Let x and y be distinct non-adjacent vertices in a graph G , and let κ' be the cardinality of a minimum x, y -edge cut. Any x, y -edge cut must contain an edge from each x, y -path. Thus, κ' is greater than or equal to the maximum number of pairwise edge-disjoint x, y -paths.

We need to show that there exist κ' pairwise edge-disjoint x, y -paths. For this we use the max-flow min-cut theorem. Construct a network N from G as follows. In the directed case, each edge is assigned capacity 1. In the undirected case, each edge uv of G is replaced with two directed edges uv and vu , each with capacity 1. The source is x and the sink is y . It is clear that the cardinality of any x, y -cut $[S, T]$ in G is equal to the capacity of the cut $[S, T]$ in N . Thus, the minimum capacity of a cut in N is κ' , and so by the max-flow min-cut theorem, N has a maximum flow of value κ' . The result follows by Theorem 3.1.11. $\square$

Chapter 4

Planar Graphs

Definition 4.0.1. A graph is **planar** if it can be drawn in the plane with no edge crossings. Such a drawing is called a **plane graph**, and the graph is said to be **embedded** in the plane.

Arguments and definitions concerning embeddings of graphs can be made rigorous and formal, but we will use just an intuitive understanding of embedding and planarity, such as in the above definition. A more formal treatment makes use of the *Jordan Curve Theorem* which (roughly speaking) states that any simple closed curve divides the plane into two regions, the interior and the exterior. A full topological proof of the Jordan Curve Theorem is quite difficult.

To illustrate the relevance of the Jordan Curve Theorem, consider whether the graph $K_{3,3}$ can be embedded in the plane. If $K_{3,3}$ is embedded in the plane, then any 6-cycle forms a simple closed curve, and hence, by the Jordan Curve Theorem, it makes sense to talk about the interior and the exterior of the 6-cycle.

Given a cycle C , we say that two chords of C are conflicting if their endpoints occur in alternating order on C . The point of conflicting chords is that it is not possible to draw both of them in the interior of C without the two chords crossing. Similarly, any two conflicting chords both drawn in the exterior of C will cross. Thus, since a 6-cycle C in $K_{3,3}$ has three pairwise conflicting chords, any drawing has at least two chords of C in the interior of C , or at least two chords of C in the exterior of C . So $K_{3,3}$ cannot be embedded in the plane.

Definition 4.0.2. The **faces** of a plane graph are the resulting connected pieces of the plane that remain if the vertices and edges are removed. One of the faces is unbounded and is called the **exterior face**. The **boundary** of a face consists of the vertices and edges that are incident with the face. Two faces are **adjacent** or **neighbouring** if they are separated by a single edge, and the **length** of a face is the number of edges on its boundary plus the number of bridges on its boundary. The length of a face f is denoted by $\text{len}(f)$.

In the above definition, the length of a face is defined to be the number of edges on the boundary, except that bridges are counted twice. Intuitively, bridges are counted twice because both sides of a bridge are on the boundary.

Definition 4.0.3. The **dual graph** of a plane graph G is the plane graph obtained from G by placing a vertex in each face of G , and then for each edge e of G , joining the vertices in the two faces on either side of e by an edge that passes through e . If the same face is on both sides of e , then a loop results. The dual of a plane graph G is denoted by G^* .

Theorem 4.0.4. For each vertex v of G^* , we have $\deg_{G^*}(v) = \text{len}(f)$ where f is the face of G corresponding to v . If F denotes the set of all faces of a plane graph G , then

$$\sum_{f \in F} \text{len}(f) = 2e(G).$$

Proof Let v be a vertex of G^* and let f be the corresponding face of G . For each edge e on the boundary of f , there is a corresponding edge e^* incident with v , and if e is a bridge, then there is a corresponding loop on v . Since loops count 2 towards the degree of a vertex, and bridges count two towards the length of a face, we have $\deg_{G^*}(v) = \text{len}(f)$.

Thus, if F denotes the set of faces of G , then we have

$$\sum_{f \in F} \text{len}(f) = \sum_{v \in V(G^*)} \deg(v) = 2e(G^*) = 2e(G).$$

□

4.1 Euler's Identity and Kuratowski's Theorem

Theorem 4.1.1. (Euler's Identity) If a connected plane graph has n vertices, m edges, and r faces, then

$$n - m + r = 2.$$

Proof The proof is by induction on m . The only connected graph with zero edges is K_1 . In this case $n = 1$, $m = 0$, $r = 1$ and $n - m + r = 2$. Thus we have $m \geq 1$ for the inductive step. Let G be a connected plane graph, let n be the number of vertices of G , let $m \geq 1$ be the number of edges of G , and let r be the number of faces of G .

If G is a tree, then $n = m + 1$ (because in any tree, the number of vertices is one more than the number of edges) and $r = 1$ so $n - m + r = 2$. If G is not a tree, then G contains an edge e that is in a cycle. The graph $G - e$ has n vertices, $m - 1$ edges and $r - 1$ faces (because there are distinct faces of G on either side of e that are part of the same face of $G - e$). Thus, by the inductive hypothesis, we have $n - (m - 1) + (r - 1) = 2$. So $n - m + r = 2$ and the result follows by induction. □

Theorem 4.1.2. If G is a simple n -vertex planar graph with $n \geq 3$, then $e(G) \leq 3n - 6$.

Proof It suffices to consider connected graphs because a new planar graph can be obtained by adding more edges to any disconnected graph. Since G is simple and has at least three vertices, every face has length at least 3. Let F be the set of faces of an embedding of G and let $r = |F|$. We have $2e(G) = \sum_{f \in F} \text{len}(f) \geq 3r$. Substituting $r \leq \frac{2}{3}e(G)$ into Euler's Identity, we obtain $e(G) \leq 3n - 6$. □

Theorem 4.1.3. The graph K_5 is non-planar.

Proof Theorem 4.1.2 tells us that a simple 5-vertex planar graph has at most $3 \times 5 - 6 = 9$ edges. Since K_5 has 10 edges it is non-planar. $\square$

We have already noted above that the graph $K_{3,3}$ is non-planar. We now state this as a theorem, and give a different proof using ideas similar to those that were used to prove Theorem 4.1.2.

Theorem 4.1.4. The graph $K_{3,3}$ is non-planar.

Proof For a contradiction, suppose $K_{3,3}$ is planar, let F be the set of faces of an embedding of $K_{3,3}$ in the plane, and let $r = |F|$. Euler's Identity tells us that $6 - 9 + r = 2$. So $r = 5$. Since $K_{3,3}$ is K_3 -free, the boundary of each face has at least 4 edges. Thus, $18 = 2e(K_{3,3}) = \sum_{f \in F} \text{len}(f) \geq 4r = 20$. We have a contradiction and conclude that $K_{3,3}$ is non-planar. $\square$

Definition 4.1.5. Any graph obtained from a graph G by replacing edges with pairwise internally disjoint paths is called a **subdivision** of G .

It can be seen that a graph G is planar if and only if every subdivision of G is planar. Thus, any graph containing a subdivision of K_5 or $K_{3,3}$ is non-planar. In 1930, Kuratowski [11] proved the converse of this.

Theorem 4.1.6. (Kuratowski's Theorem) A graph is planar if and only if it does not contain a subdivision of K_5 or $K_{3,3}$.

Proof Omitted. There are proofs in [3] and [22], and in many other graph theory texts. $\square$

4.2 The Four Colour Theorem

The *Four Colour Problem* asks ...

Is it true that the countries of any map can be coloured with four colours so that any two neighbouring countries are assigned distinct colours?

The *Four Colour Theorem* states that four colours are indeed sufficient.

The Four Colour Problem originated in England in the early 1850s. It was formulated by Francis Guthrie who observed that four colours were sufficient to colour the counties of England. The problem was passed on to Augustus De Morgan, Professor of Mathematics at University College in London, by Francis' elder brother Frederick, who was a student of De Morgan's at the time. The earliest record of the problem is a letter written by De Morgan to William Rowan Hamilton, the famous mathematician and physicist, on 23rd October, 1852. The problem first appeared in print on 14th April, 1860.

In 1880, Alfred Bray Kempe published what is now known to be an incorrect proof of the Four Colour Theorem. This is perhaps the most famous "false proof" in all of mathematics. The Four

Colour Problem is notorious for generating numerous “proofs” which appear convincing to some at first, but are fatally flawed. The problem has attracted the interest of many amateur mathematicians over the years. The error in Kempe’s proof was not found until ten years later, in 1890, by Percy Heawood. Although Kempe’s proof was ultimately incorrect, his work introduced very useful ideas and techniques for attacking the problem.

Kenneth Appel and Wolfgang Haken, both from the University of Illinois, announced their proof of the Four Colour Theorem, on 22nd July, 1976. It was published in a paper in the Illinois Journal of Mathematics in 1977 [1]. The extensive use of computers in their proof generated a great deal of scepticism, perhaps initially bordering on rejection. It also generated a certain amount of disappointment. Their proof was undoubtedly a great piece of work, but it was long, tedious and complicated, rather than elegant and intuitive. In 1981, Ulrich Schmidt found an error in the proof, but this was quickly corrected. In 1997, Neil Robertson, Daniel Sanders, Paul Seymour and Robin Thomas published a new proof of the Four Colour Theorem. These proofs are now widely accepted in the mathematical community. However, there is still an amount of scepticism, or at least uneasiness amongst some mathematicians.

Before proceeding further, we need a few words of explanation (but not an entirely rigorous and formal definition) so as to make sure that we understand the Four Colour Problem clearly, and what the underlying assumptions are.

A map consists of a finite number of regions of the plane. The *regions* may represent countries, states, counties, etc, but can also be entirely fictitious. Each region must be connected (consist of a single piece of the plane).

Two regions are *adjacent* or *neighbouring* if they share a common boundary. However, two regions meeting at a single point are not adjacent. If such regions were considered adjacent, then there would be maps that require arbitrarily many colours (if neighbouring regions are to be assigned distinct colours). For similar reasons, the restriction that each region be a single connected piece of the plane is important.

It is apparent that any map has a corresponding plane graph G such that regions of the map are adjacent if and only if the corresponding faces of G are adjacent. Thus, the Four Colour Problem is equivalent to asking whether it is possible to colour the faces of any plane graph with four colours so that adjacent faces are assigned distinct colours. By considering the dual graph G^* of G , this becomes a problem concerning vertex colourings of planar graphs.

Thus, the Four Colour Problem is equivalent to asking whether every planar graph is 4-colourable. In 1890, Heawood proved that every planar graph is 5-colourable [8].

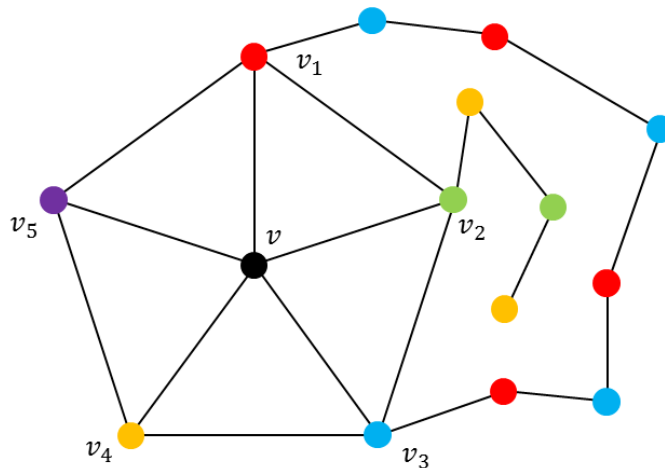
Theorem 4.2.1. (Five Colour Theorem) Every planar graph is 5-colourable.

Proof Suppose the theorem is false and let G be a counter-example of minimum order. Let n be the order of G . Every graph with 5 or fewer vertices is 5-colourable, so $n \geq 6$. If every vertex of G has degree at least 6, then G has at least $3n$ edges. But this contradicts Theorem 4.1.2 and we conclude that G has a vertex v of degree at most 5.

The graph $G - v$ is 5-colourable by the minimality of G , so we can give $G - v$ a proper 5-colouring. Each of the 5 colours must occur on a neighbour of v because otherwise there is a colour available

for vertex v . Thus, v has degree 5 and the 5 neighbours of v are assigned distinct colours. Let v_1, v_2, v_3, v_4, v_5 be the five neighbours of v in clockwise order around v , and without loss of generality for $i = 1, 2, 3, 4, 5$, let i be the colour of v_i .

Let $G_{1,3}$ be the subgraph of $G - v$ induced by the vertices of colour 1 and 3. If v_1 and v_3 are in different components of $G_{1,3}$, then we can switch the colours on the component of $G_{1,3}$ that contains v_1 and then produce a proper 5-colouring of G by assigning colour 1 to v . Thus, v_1 and v_3 are in the same component of $G_{1,3}$, and there exists a v_1, v_3 -path all of whose vertices are coloured 1 or 3. This path can be completed to a cycle C by adding the vertex v and the edges vv_1 and vv_3 .



One of the vertices v_2 and v_4 is in the interior of C and the other is in the exterior. Thus, v_2 and v_4 are in different components of $G_{2,4}$, where $G_{2,4}$ is the subgraph of $G - v$ induced by the vertices of colour 2 and 4. Thus, if we switch the colours on the component of $G_{2,4}$ that contains v_2 and assign colour 2 to v , then we obtain a proper 5-colouring of G . This is a contradiction and we conclude that every planar graph is 5-colourable. $\square$

Definition 4.2.2. A planar graph in which the boundary of every face is a triangle is called a **planar triangulation**.

Theorem 4.2.3. Every planar graph is 4-colourable if and only if every planar triangulation is 4-colourable.

Proof If every planar graph is 4-colourable, then certainly every planar triangulation is 4-colourable. To prove the converse, suppose that every planar triangulation is 4-colourable and let G be a plane graph. If G is not a triangulation, then we can add edges to G until we obtain a planar triangulation. By assumption, this planar triangulation has a proper 4-colouring, and this is also a proper 4-colouring of G . Thus, G is 4-colourable. $\square$

Definition 4.2.4. In the context of the Four Colour Problem, a **minimal counter-example** is a planar triangulation G such that G is not 4-colourable, and such that every planar triangulation with fewer vertices than G is 4-colourable.

Definition 4.2.5. A **separating cycle** in a planar triangulation is a cycle C such that if the vertices of C together with any edges incident with any of them are deleted, then the resulting graph has at least two connected components.

Definition 4.2.6. A **configuration** in a planar triangulation consists of a separating cycle C , called the **ring**, together with the part of the graph inside C . A set S of configurations is **unavoidable** if any minimal counter-example must contain a copy of at least one configuration from S . A configuration is **reducible** if a planar triangulation containing it cannot be a minimal counter-example.

By definition, if there exists an unavoidable set of reducible configurations, then every planar graph is 4-colourable. That is, to prove that every planar graph is 4-colourable, it is sufficient to find an unavoidable set of reducible configurations.

One method for proving that sets of configurations are unavoidable is known as *discharging*. It is based on the following theorem.

Theorem 4.2.7. If G is a planar triangulation, then

$$\sum_{v \in V(G)} (6 - \deg(v)) = 12.$$

Proof Let F be the set of faces of a planar triangulation and let $r = |F|$. Then

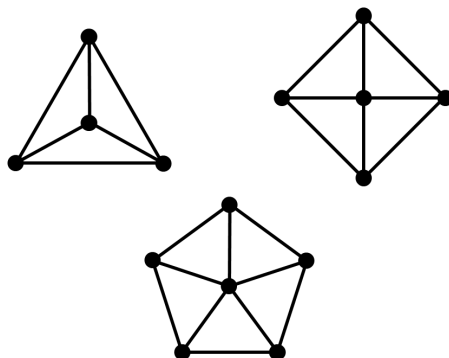
$$\sum_{v \in V(G)} \deg(v) = 2e(G) = \sum_{f \in F} \text{len}(f) = 3r.$$

But by Euler's Identity we have $6n(G) - 6e(G) + 6r = 12$, so

$$12 = 6n(G) - 3 \sum_{v \in V(G)} \deg(v) + 2 \sum_{v \in V(G)} \deg(v) = 6n(G) - \sum_{v \in V(G)} \deg(v) = \sum_{v \in V(G)} (6 - \deg(v)).$$

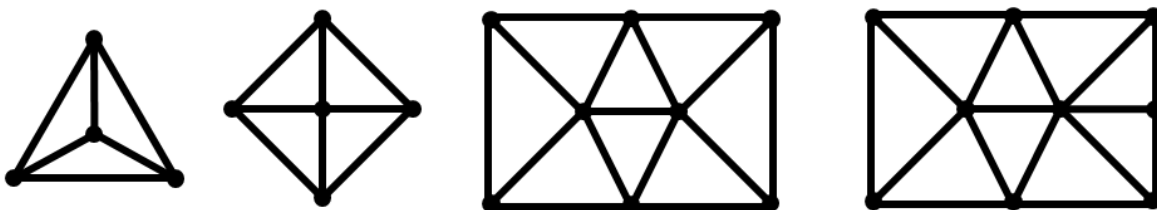
□

If every vertex in a planar triangulation has degree at least 6, then every term in the sum on the left of the expression in Theorem 4.2.7 would be at most zero, a contradiction. Thus, every planar triangulation has a vertex of degree at most 5 (a fact we already noted in the proof of the Five Colour Theorem, Theorem 4.2.1). That is, the following three configurations form an unavoidable set.



A more interesting use of discharging is illustrated in the proof of the following theorem.

Theorem 4.2.8. The following set of configurations is unavoidable.



Proof For a contradiction, suppose there is a planar triangulation G containing none of the four given configurations. Thus, there are no vertices of degree 3 or 4, and the neighbours of any vertex of degree 5 have degree at least 7. (It can be seen that if there is a vertex of degree 3 or 4, or a vertex of degree 5 with a neighbour of degree 5 or 6, then at least one of these configurations occurs – in the two rightmost configurations, the vertices on the rings must be distinct as drawn in order to avoid a vertex of degree 3 or 4.) Define the *charge* on each vertex v to be $6 - \deg(v)$. Then Theorem 4.2.7 says that the sum of the vertex charges, the *net charge*, in G is 12.

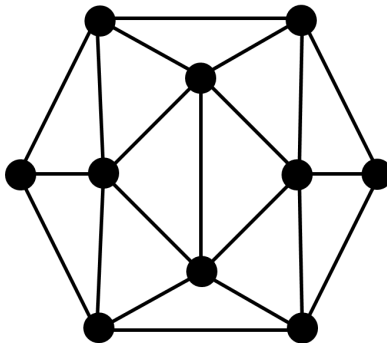
Redistribute the charge by shifting $\frac{1}{5}$ of the charge on each vertex v of degree 5 to each of the 5 neighbours of v . The net charge is unchanged and each vertex of degree 5 now has charge zero. Each vertex of degree 6 still has charge zero.

Each vertex v of degree $d \geq 7$ now has charge $(6 - d) + \frac{1}{5}t(v)$, where $t(v)$ is the number of neighbours of v having degree 5. Since $t(v) \leq d$, each vertex of degree at least 8 still has negative charge, and each vertex of degree 7 has charge at most zero, unless it has at least 6 neighbours of degree 5. But a vertex of degree 7 cannot have more than 3 neighbours of degree 5 because no two vertices of degree 5 are adjacent.

Thus, the net charge is at most zero, a contradiction. We conclude that the set of configurations is unavoidable. $\square$

Throughout the early to mid 1900s, methods of increasing complexity were used to construct reducible configurations and unavoidable sets.

In 1913, George Birkhoff, who is famous in many areas of mathematics and particularly well known for his work on ergodic theory and the three-body problem, proved that any configuration with at least seven vertices whose ring is a 5-cycle is reducible. He also proved that what is now known as the *Birkhoff diamond* (shown below) is reducible.



Proofs of reducibility of configurations are long and complicated. One of the difficulties is the rapid increase in the number of distinct possible colourings of the ring as the size (number of edges) of the ring increases. Each of these colourings needs to be considered in reducibility proofs. The number of distinct colourings for rings of sizes 6 to 14 is given in the table below [25].

ring size	6	7	8	9	10	11	12	13	14
number of colourings	31	91	274	820	2461	7381	22144	64430	199291

By the end of the 1960s, Haken and others had been using computers extensively to help with the huge number of cases that needed to be considered in the *unavoidable set of reducible configurations* approach to the Four Colour Problem. In late June, they constructed an unavoidable set of 1936 configurations with ring sizes at most 14. A couple of days later they had proved that the set was also reducible.

In the 1977 published version of the Appel and Haken's proof [1], the number of reducible configurations in their unavoidable set had been reduced from 1936 to 1482. The later proof of Robertson, Sanders, Seymour and Thomas [14], which was published in 1997, used an unavoidable set of 633 configurations.

4.3 Snarks

We have seen (see Theorem 4.2.3) that the Four Colour Theorem is equivalent to the statement that every planar triangulation is 4-colourable. The dual of a planar triangulation is a simple 2-edge-connected 3-regular planar graph. Thus, the Four Colour Theorem is also equivalent to the statement that the faces of every simple 2-edge-connected 3-regular planar graph can be coloured with 4 colours so that adjacent faces are assigned distinct colours.

We have similar definitions for face colourings of plane graphs as we have for vertex and edge colourings of graphs.

Definition 4.3.1. A **face colouring** of a plane graph is an assignment of colours to its faces, and a face colouring with k or fewer colours is a **k -face colouring**. A k -face colouring is **proper** if adjacent faces are assigned distinct colours, and a plane graph is **k -face-colourable** if it has a proper k -face colouring.

The following result, which was proved by Tait in 1878 [16], shows that the Four Colour Theorem is also equivalent to the statement that every simple 2-edge-connected 3-regular plane graph is 3-edge-colourable.

Theorem 4.3.2. Let G be a simple 2-edge-connected 3-regular planar graph. Then G is 4-face-colourable if and only if G is 3-edge-colourable.

Proof First suppose that G has a proper 4-face colouring. Let the four colours be 00, 01, 10 and 11, the elements of the elementary abelian 2-group $\mathbb{Z}_2 \times \mathbb{Z}_2$. Colour each edge e of G with the sum of the colours of the faces that e separates. We claim that this produces a proper 3-edge colouring of G with colours 01, 10 and 11.

To see this, first observe that 00 never occurs as each edge separates faces of distinct colours. If the colours on the pairwise adjacent faces around a vertex v are 01, 10 and 11, then it follows that these same 3 colours occur on the three edges incident with v (because $01 + 10 = 11$, $01 + 11 = 10$ and $10 + 11 = 01$). If the colours on the pairwise adjacent faces around a vertex v are 00, c_1 and c_2 , where $c_1, c_2 \in \{01, 10, 11\}$, then it follows that the two edges incident with v and on the boundary of the face coloured 00 are assigned colours c_1 and c_2 and the third edge incident with v is assigned colour $c_1 + c_2$, which is the third element of $\{01, 10, 11\}$. Thus, we have a proper 3-edge colouring of G .

For the converse, now suppose that G has a proper 3-edge colouring. Let these colours be red, blue and green. Since G is 3-regular, the red and blue edges induce a 2-factor $G_{red-blue}$ of G . Similarly, the blue and green edges also induce a 2-factor $G_{blue-green}$ of G . Note that a 2-regular graph is the vertex disjoint union of cycles.

We assign colours from $\{00, 01, 10, 11\}$ to the faces of G as follows. The first coordinate of the colour of a face is the parity (0 for even, 1 for odd) of the number of cycles of $G_{red-blue}$ containing the face, and the second coordinate is the parity of the number of cycles of $G_{blue-green}$ containing the face. We claim that this is a proper face colouring of G .

To see this, consider two neighbouring faces and let e be the edge that separates them. If e is red, then exactly one of the faces adjacent to e is in the interior of the cycle of $G_{red-blue}$ that contains e , and no other cycle of $G_{red-blue}$ separates the two faces. Thus, the number of cycles of $G_{red-blue}$ that contain one of the faces adjacent to e differs from the number containing the other by 1. Hence the colours the two faces are assigned differ in their first coordinates.

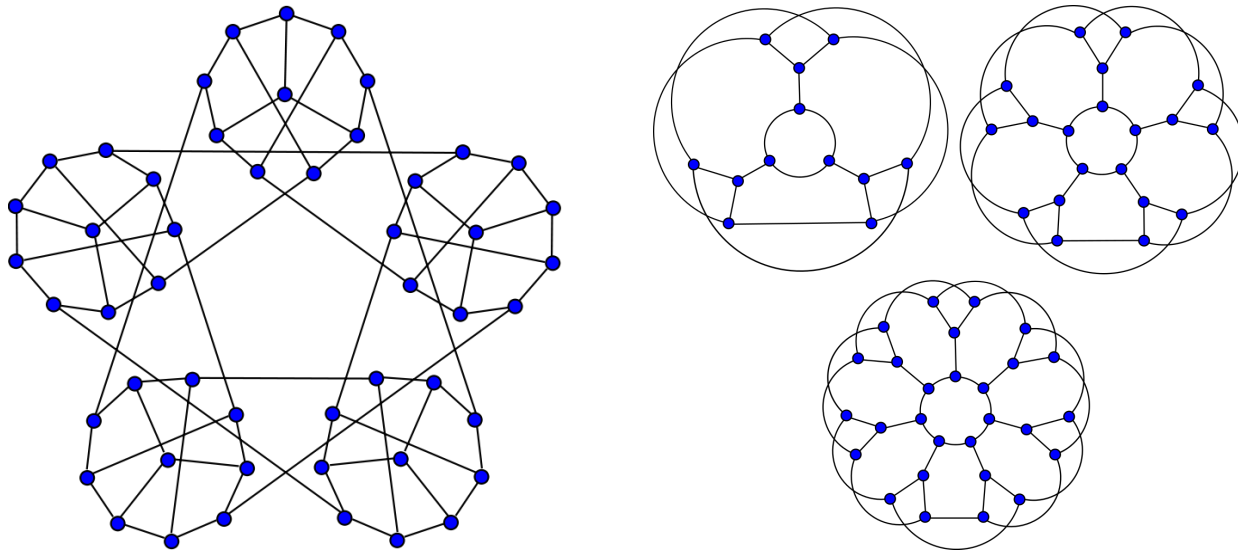
By similar arguments, if two neighbouring faces are separated by a green edge, then the colours they are assigned differ in the second coordinates, and if the two neighbouring faces are separated by a blue edge, then the colours they are assigned differ in both coordinates. Thus, we do indeed have a proper 4-face colouring of G . $\square$

Since the Four Colour Theorem is equivalent to the statement that every simple 2-edge-connected 3-regular planar graph is 3-edge-colourable, simple 2-edge-connected 3-regular graphs that are not 3-edge-colourable (these are necessarily non-planar by the Four Colour Theorem) are of interest. Such graphs are called *snarks*.

Definition 4.3.3. A **snark** is a simple 2-edge-connected 3-regular graph that is not 3-edge-colourable.

If G is a snark and we replace some vertex v of G by a triangle (so v is removed and the three edges incident on v are joined instead to the three vertices of the triangle), then we obtain another snark. There are other ways of constructing new snarks from existing snarks, and as a consequence sometimes more conditions must be satisfied in order for a graph to be called as a snark. For example, snarks are sometimes required to have girth at least 5 and to have no non-trivial 3-edge cuts.

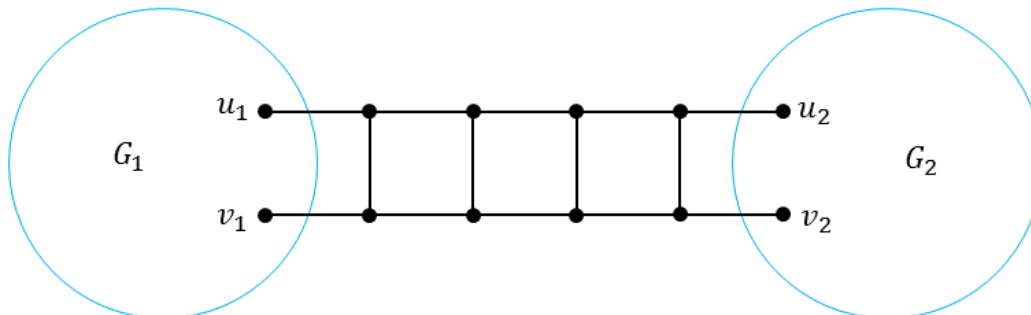
The Petersen graph is the smallest snark. Shown below is the Szekeres snark (at left) and the Flower snarks J_3 , J_5 and J_7 (at right).



4.4 Grinberg's Theorem and the Tutte Graph

Definition 4.4.1. Let G be a connected 3-regular graph and let u_1, u_2, v_1, v_2 be distinct vertices of G . A **ladder of length $t + 1$ in G joining (u_1, v_1) to (u_2, v_2)** consists of disjoint paths $[u_1, u'_1, u'_2, \dots, u'_t, u_2]$ and $[v_1, v'_1, v'_2, \dots, v'_t, v_2]$ and edges $u'_1v'_1, u'_2v'_2, \dots, u'_tv'_t$ such that $\{u_1u'_1, v_1v'_1\}$ is an edge cut of G .

Lemma 4.4.2. Let G be a 3-regular graph with edge connectivity 2. Then G consists of vertex-disjoint subgraphs G_1 and G_2 , non-adjacent vertices $u_1, v_1 \in V(G_1)$, non-adjacent vertices $u_2, v_2 \in V(G_2)$, and a ladder joining (u_1, v_1) to (u_2, v_2) .



Proof If G has a 2-edge cut consisting of incident edges, then the third edge on the vertex where they are incident is a bridge, contradicting the fact that G has edge connectivity 2. Thus, there exists a 2-edge cut $\{u_1u'_1, v_1v'_1\}$ where u_1, v_1 are distinct and u'_1, v'_1 are distinct. If u_1, v_1 and u'_1, v'_1 are non-adjacent, then we have the desired ladder of length 1 consisting of the paths $[u_1, u'_1 = u_2]$ and $[v_1, v'_1 = v_2]$.

If u_1v_1 are non-adjacent and u'_1, v'_1 are adjacent, then we let u'_2 be the third neighbour of u'_1 (the other two neighbours being u_1 and v'_1) and let v'_2 be the third neighbour of v'_1 (the other two neighbours being v_1 and u'_1). If $u'_2 = v'_2$, then the third edge incident on this vertex is a bridge. So u'_2 and v'_2 are distinct. If u_1, v_1 and u'_2, v'_2 are non-adjacent, then we have the desired ladder of length 2 consisting of the paths $[u_1, u'_1, u'_2 = u_2]$ and $[v_1, v'_1, v'_2 = v_2]$ and the edge $u'_1v'_1$.

If u_1, v_1 are non-adjacent and u'_2, v'_2 are adjacent, then we let u'_3 be the third neighbour of u'_2 (the other two neighbours being u'_1 and v'_2) and let v'_3 be the third neighbour of v'_2 (the other two neighbours being v'_1 and u'_2), and continue “extending the ladder” until we reach non-adjacent vertices u'_{t+1} and v'_{t+1} , so that we have the required ladder consisting of disjoint paths $[u_1, u'_1, u'_2, \dots, u'_t, u'_{t+1} = u_2]$ and $[v_1, v'_1, v'_2, \dots, v'_t, v'_{t+1} = v_2]$ and the edges $u'_1v'_1, u'_2v'_2, \dots, u'_tv'_t$.

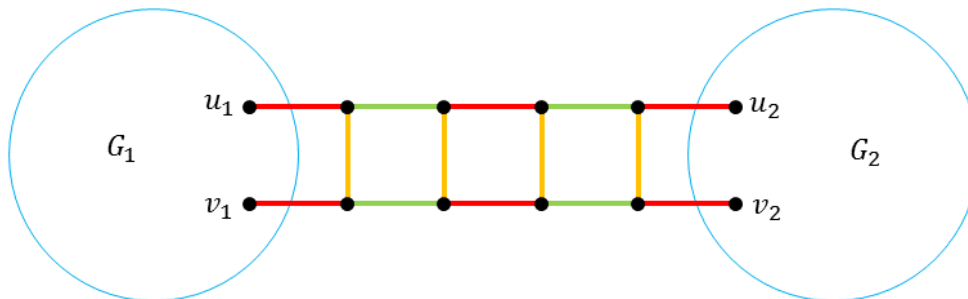
We can repeat the same process “going in the other direction” if u_1, v_1 are adjacent. □

Theorem 4.4.3. Every 2-edge-connected 3-regular simple planar graph is 3-edge-colourable if and only if every 3-edge-connected 3-regular simple planar graph is 3-edge-colourable.

Proof Since every 3-edge-connected graph is also 2-edge-connected, we only need to show that if every 3-edge-connected 3-regular simple planar graph is 3-edge-colourable, then every 2-edge-connected 3-regular simple planar graph is 3-edge-colourable. So suppose that every 3-edge-connected 3-regular simple planar graph is 3-edge-colourable, and let G be a 2-edge-connected 3-regular simple planar graph. We need to show that G is 3-edge-colourable. The proof of this is by induction on $n(G)$.

For $n(G) \leq 4$, the only graph to consider is K_4 , which is 3-edge-colourable. Thus, for the inductive step we have $n(G) > 4$. If G is 3-edge-connected, then it is 3-edge-colourable by assumption. So we can assume that G has edge connectivity 2. Thus, by Lemma 4.4.2, G consists of vertex-disjoint subgraphs G_1 and G_2 , non-adjacent vertices $u_1, v_1 \in V(G_1)$, non-adjacent vertices $u_2, v_2 \in V(G_2)$, and a ladder joining (u_1, v_1) to (u_2, v_2) . Let the ladder consist of the paths $[u_1, u'_1, u'_2, \dots, u'_t, u'_{t+1} = u_2]$ and $[v_1, v'_1, v'_2, \dots, v'_t, v'_{t+1} = v_2]$ and the edges $u'_1v'_1, u'_2v'_2, \dots, u'_tv'_t$.

Both $G_1 + u_1v_1$ and $G_2 + u_2v_2$ are 2-edge-connected and 3-regular. So by the inductive hypothesis they are 3-edge-colourable. Colour the rungs (edges $u'_1v'_1, u'_2v'_2, \dots, u'_tv'_t$) of the ladder with colour c_3 and alternately colour the edges of the paths $[u_1, u'_1, u'_2, \dots, u'_t, u'_{t+1} = u_2]$ and $[v_1, v'_1, v'_2, \dots, v'_t, v'_{t+1} = v_2]$ of the ladder with c_1 and c_2 , beginning by colouring the edges $u_1u'_1$ and $v_1v'_1$ with colour c_1 . Thus, the ladder is properly 3-edge-coloured.



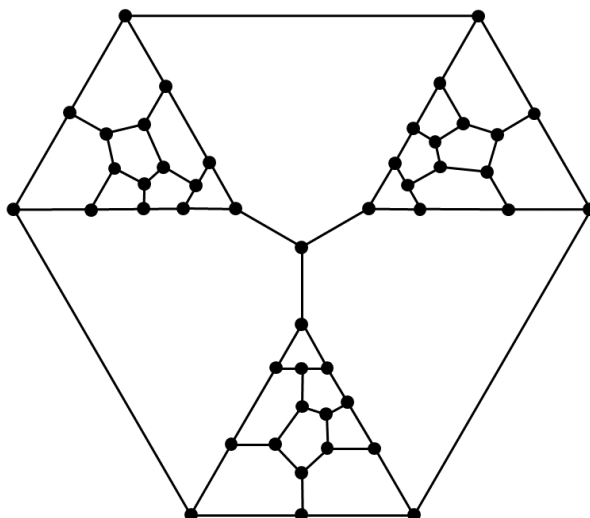
By permuting the colours in a proper 3-edge colouring of $G_1 + u_1v_1$ with colours c_1, c_2, c_3 , we can ensure that the edge u_1v_1 is assigned colour c_1 . Thus, the union of G_1 and the ladder is properly 3-edge-coloured. Similarly, by permuting the colours in a proper 3-edge colouring of $G_2 + u_nv_n$ with colours c_1, c_2, c_3 , we can ensure that the edge u_nv_n is assigned the same colour as the edges $u'_t u_2$ and $v'_t v_2$ (namely colour c_1 if t is even, and colour c_2 if t is odd). This produces the required proper 3-edge colouring of G . $\square$

Theorems 4.3.2 and 4.4.3 tell us that the Four Colour Theorem is equivalent to the statement that every simple 3-edge-connected 3-regular planar graph is 3-edge-colourable. We also know that a 3-regular graph (with more than two vertices) is 3-edge-connected if and only if it is 3-connected (Theorem 3.0.5). So the Four Colour Theorem is also equivalent to the statement that every simple 3-connected 3-regular planar graph is 3-edge-colourable.

In 1880, Tait conjectured that every simple 3-connected 3-regular planar graph is Hamiltonian. It is easily seen that a Hamiltonian 3-regular graph is 3-edge-colourable. So Tait's conjecture, if true, implies the Four Colour Theorem.

It wasn't until 1946, that a counter-example to Tait's conjecture was found. This counter-example was found by Tutte [17] and is now known as the *Tutte graph*.

Definition 4.4.4. The **Tutte graph** is the graph shown below.



The following theorem of Grinberg [7] can be used to prove that the Tutte graph has no Hamilton cycle.

Theorem 4.4.5. (Grinberg's Theorem) Let G be a loopless plane graph with a Hamilton cycle C , let r_i denote the number of faces of length i in the interior of C , and let r'_i denote the number of faces of length i in the exterior of C . Then

$$\sum_i r_i(i-2) = \sum_i r'_i(i-2) = n(G) - 2.$$

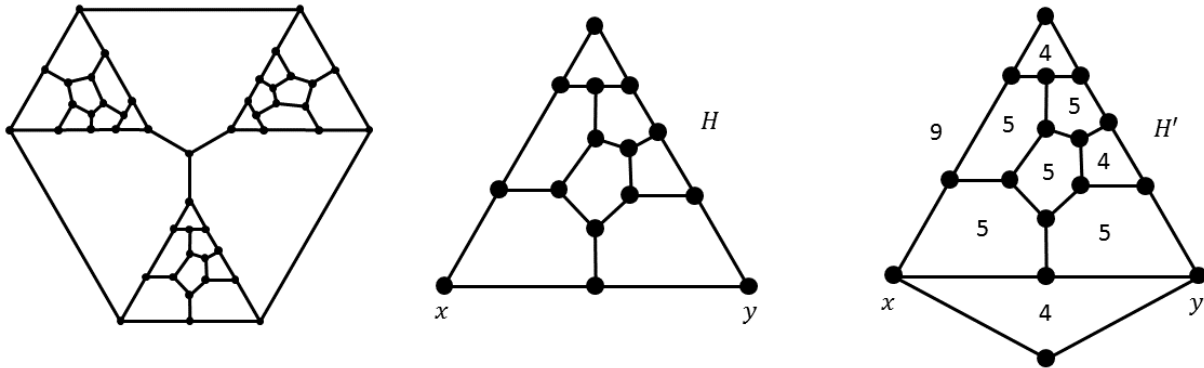
Proof We begin by showing that $\sum_i(i-2)r_i = n(G) - 2$. The proof of this is by induction on the number t of edges in the interior of C . If there are no edges in the interior of C , then the result holds. Thus, we have $t \geq 1$ for the inductive step.

Let G' be the graph obtained from G by deleting an edge e from the interior of C , and suppose that in G the two faces on either side of e have lengths a and b . In G' , instead of these two faces we have a single face of length $a + b - 2$ (and all the other faces are unchanged). Thus, by the inductive hypothesis, and since $(a-2) + (b-2) = (a+b-2) - 2$, we have $\sum_i(i-2)r_i = n(G) - 2$.

There is no fundamental difference between the interior and the exterior of C (consider an embedding on a sphere rather than in the plane). So we also have $\sum_i(i-2)r'_i = n(G) - 2$. $\square$

Theorem 4.4.6. The Tutte graph is non-Hamiltonian.

Proof For a contradiction, suppose C is a Hamilton cycle in the Tutte graph. Consider the three isomorphic components that remain if the central vertex and the three longest edges of the Tutte graph (as shown below) are deleted. Let H be the component that does not contain an endpoint of the two edges of C that are incident with the central vertex. The edges of C that are in H form a Hamilton x, y -path in H where x and y are the two vertices as indicated below. Thus, the graph H' obtained from H by adding a new vertex that is joined to x and y has a Hamilton cycle C' .



The graph H' has 16 vertices, five faces of length 5, three faces of length 4 and one face of length 9 (the exterior face). The two faces of length 4 that are incident with a vertex of degree 2 are both in the interior of C' . Let x be the number of faces of length 5 in the exterior of C' . Applying Grinberg's Theorem to the exterior cycles, we have

$$7 + 3x = 14 \quad \rightarrow \quad 3x = 7$$

if the third face of length 4 is in the interior of C' , or

$$7 + 3x + 2 = 14 \quad \rightarrow \quad 3x = 5$$

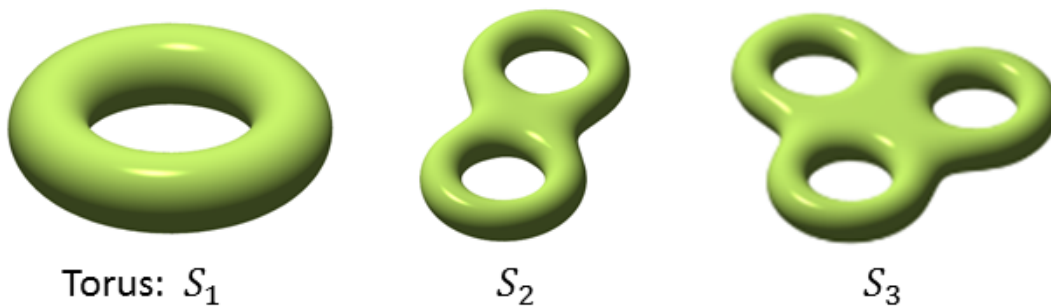
if the third face of length 4 is in the exterior of C' . This is a contradiction and we conclude that the Tutte Graph is non-Hamiltonian. $\square$

4.5 Embeddings on other Surfaces

An embedding of a graph on an arbitrary surface is defined in a similar way as an embedding of a graph in the plane; the graph is drawn on the surface such that there are no edge-crossings.

A graph can be embedded in the plane if and only if it can be embedded on a sphere. To see this imagine puncturing a face of a graph that is embedded on the sphere and then stretching the surface out flat. The punctured face becomes the external face. This process is reversible. We shall see however that there are non-planar graphs that can be embedded on other surfaces.

A **torus**, the familiar “doughnut-shaped” surface, can be constructed in several different ways. One way is to begin with the unit square, identify the point $(x, 0)$ with the point $(x, 1)$ for each x in the interval $[0, 1]$, and identify the point $(0, y)$ with the point $(1, y)$ for each y in the interval $[0, 1]$. Another way to construct a torus is to glue a handle to a sphere. Generalising this construction, the surface obtained from a sphere by the addition of k handles is the **orientable surface of genus k** , and is denoted by S_k .



Any graph can be drawn on a sphere such that there is no point where more than two edges cross. At any point where two edges do cross, a handle can be placed so that one of the edges goes around the handle and the crossing is avoided. Thus, any graph can be embedded on some orientable surface. For a given graph G , the smallest non-negative integer k such that G can be embedded on S_k is called the **genus** of G .

A face f of a graph that is embedded on a surface is a **2-cell** if every closed curve in f can be continuously deformed within that region to a point (that is, a face is a 2-cell if it is homeomorphic to a disk). An embedding of a graph on a surface is a **2-cell embedding** if every face of the embedding is a 2-cell. It can be seen that any embedding of a connected graph on a sphere is a 2-cell embedding. This is not true for other surfaces.

The following theorem generalises Euler's Identity.

Theorem 4.5.1. If a connected graph having n vertices and m edges has a 2-cell embedding on S_k , then the number r of faces satisfies

$$n - m + r = 2 - 2k.$$

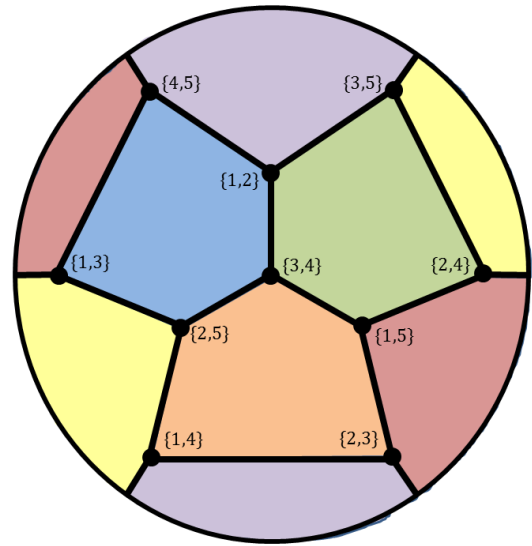
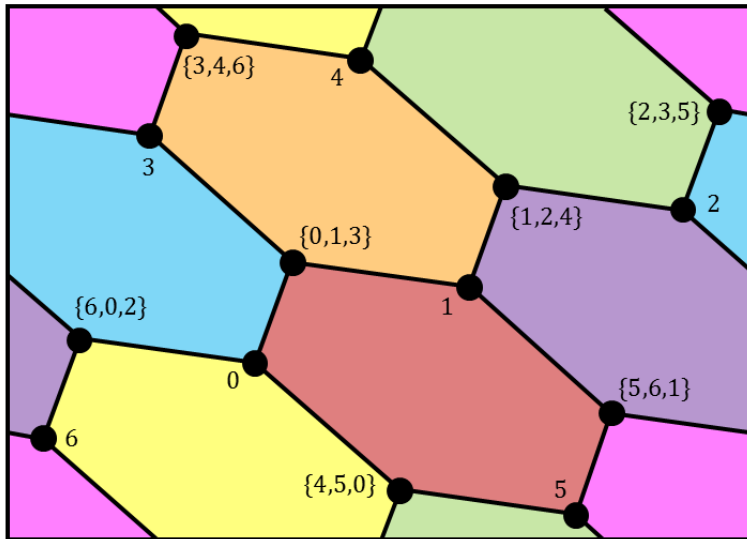
Theorem 4.5.2. If a graph of genus k is embedded on S_k , then the embedding is a 2-cell embedding.

The *projective plane* is another surface on which one can embed graphs. The **projective plane** can be constructed from the unit square by identifying the point $(x, 0)$ with the point $(1 - x, 1)$ for each x in the interval $[0, 1]$, and identifying the point $(0, y)$ with the point $(1, 1 - y)$ for each y in the interval $[0, 1]$. Alternatively, the projective plane is obtained from a circular disk by identifying each pair of antipodal points on the circumference.

Theorem 4.5.3. If a connected graph having n vertices and m edges has a 2-cell embedding on the projective plane, then the number r of faces satisfies

$$n - m + r = 1.$$

Shown below is an embedding of the Heawood graph on the torus and an embedding of the Petersen graph on the projective plane. The dual graph of this embedding of the Heawood graph is K_7 and the dual graph of this embedding of the Petersen graph is K_6 .



Chapter 5

Factors and Factorisations

5.1 Tutte's 1-Factor Theorem

The following theorem was proved by Tutte in 1947 [18].

Theorem 5.1.1. (Tutte's 1-Factor Theorem) A graph G has a 1-factor if and only if for every $S \subset V(G)$ the number of components of odd order in $G - S$ does not exceed $|S|$.

Proof Let $o(G)$ denote the number of components of odd order in a graph G . So we need to show that a graph G has a 1-factor if and only if $o(G - S) \leq |S|$ for every $S \subset V(G)$. We only need to prove the theorem for simple graphs because multiple edges make no difference to the existence of a 1-factor, nor to the condition that $o(G - S) \leq |S|$ for every $S \subset V(G)$.

If G has a 1-factor F , then for any component H of odd order in $G - S$, there is an edge of F joining a vertex of H to a vertex in S . It follows that $o(G - S) \leq |S|$.

We now proceed to show that if $o(G - S) \leq |S|$ for every $S \subset V(G)$, then G has a 1-factor. For a contradiction, suppose that there exists a counterexample G_0 to this statement. That is, G_0 is a graph such that $o(G_0 - S) \leq |S|$ for every $S \subset V(G_0)$, but G_0 has no 1-factor. Taking $S = \emptyset$, we see that G_0 has even order. Clearly, complete graphs of even order have 1-factors so G_0 is not complete.

If there exist non-adjacent vertices x and y in G_0 such that $G_0 + xy$ has no 1-factor, then $G_0 + xy$ is another counterexample. This is because joining non-adjacent vertices of a graph does not increase the number of odd components. So $G_1 = G_0 + xy$ satisfies $o(G_1 - S) \leq |S|$ for every $S \subset V(G_1)$ and G_1 has no 1-factor. That is, G_1 is also a counterexample. Thus, there exists a counter-example G such that if x and y are any non-adjacent vertices in G , then $G + xy$ has a 1-factor.

Let U be the set of vertices of G that have degree $n(G) - 1$ (so a vertex in U is adjacent to every vertex of G , except itself). The proof now splits into two cases: the case where every component of $G - U$ is complete, and the case where there is a non-complete component in $G - U$.

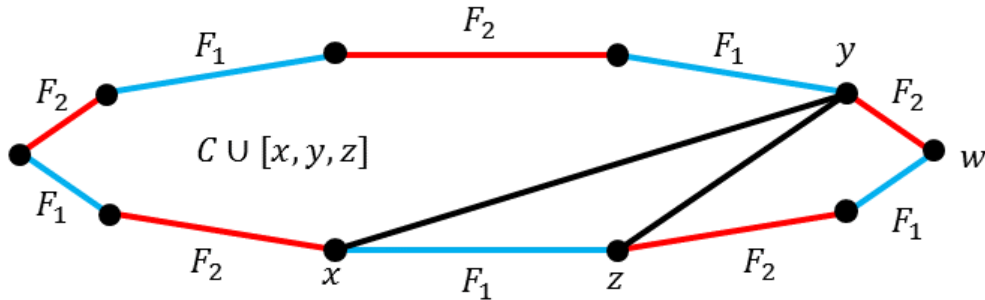
Suppose first that every component of $G - U$ is complete. Any complete graph of even order has a 1-factor, and any complete graph of odd order has a 1-regular subgraph that contains every vertex except one. Thus, there is a 1-regular subgraph F_1 of $G - U$ that contains all the vertices of $G - U$ except for one vertex from each odd-order component of $G - U$.

Since $o(G - U) \leq |U|$ and since every vertex in U is adjacent to every vertex of G (except itself), there is a 1-regular subgraph F_2 of G containing the $o(G - U)$ vertices of $G - U$ that are not in F_1 , and containing $o(G - U)$ vertices from U . Since $n(G)$ and $n(F_1 \cup F_2)$ are both even, $|V(G) \setminus V(F_1 \cup F_2)|$ is even, and by the definition of U the vertices in $V(G) \setminus V(F_1 \cup F_2)$ are pairwise adjacent. Thus, $F_1 \cup F_2$ can be extended to a 1-factor of G .

Now suppose that there is a non-complete component in $G - U$. Thus, $G - U$ contains vertices x and z that are at distance 2 (any non-complete connected graph contains non-adjacent vertices, which are at distance at least 2, and hence contains non-adjacent vertices at distance exactly 2). Let $[x, y, z]$ be a path in $G - U$. Since $y \notin U$, there exists another vertex w in $G - U$ such that y is not adjacent to w .

By the definition of G , the graphs $G + xz$ and $G + yw$ each have 1-factors. Let F_1 be a 1-factor in $G + xz$ and let F_2 be a 1-factor in $G + yw$. Thus, xy and yz are not in F_1 , and xy and yz are not in F_2 . Let F be the subgraph of G induced by the edges in exactly one F_1 and F_2 . Note that xz and yw are edges of F and that each component of F is a cycle of even length. Let C be the cycle of F that contains the edge xz .

If C does not contain yw , then the required 1-factor of G consists of the edges in F_2 that are in C and all the edges of F_1 that are not in C . Thus, we can assume that xz and yw are both in C . It is easy to see that $C \cup [x, y, z]$ has a 1-factor that contains neither of the edges xz nor yw . The edges of this 1-factor together with the edges of F_1 that are not in C form a 1-factor in G . This is a contradiction and the proof is complete. $\square$



5.2 Flows in Graphs

Definition 5.2.1. An **orientation** of a graph G is a digraph obtained from G by assigning an orientation, either x to y or y to x , to each edge xy of G .

Definition 5.2.2. A **flow** on a graph G is a pair (D, f) where D is an orientation of G and f is an assignment of weights to the edges of D such that for each vertex $v \in V(G)$, the sum of the weights on the edges directed into v equals the sum of the weights on the edges directed out of v . The weight assigned to an edge may also be referred to as the *flow* on that edge.

Note that flow is defined differently from in Section 3.1 in this section. In Section 3.1 there is a given digraph and a flow is defined to be any assignment of weights to edges. In this section there is

a given undirected graph, and a flow consists of both an orientation of the edges and an assignment of weights to edges such that “flow is conserved” at each vertex.

Lemma 5.2.3. For any flow (D, f) on a graph G , the net flow out of any subset $S \subseteq V(G)$ is zero, where the net flow out of S means the sum of the flows on the edges leaving S minus the sum of the flows on the edges entering S .

Proof By Lemma 3.1.3 the net flow out of S is the sum of the net flows out of the vertices of S , and the net flow out of each vertex is zero by definition. $\square$

Definition 5.2.4. Let $k \geq 2$ be an integer. A **nowhere-zero flow** is a flow in which no weight is zero, and a **nowhere-zero k -flow** is a flow in which all of the weights are in $\{1, 2, \dots, k-1\}$.

Theorem 5.2.5. If a graph has a nowhere-zero flow, then it has no bridges.

Proof Let $e = uv$ be a bridge in a graph G and suppose there is a flow on G . By Lemma 5.2.3, there is zero net flow out of the vertex set of the component of $G - \{e\}$ containing u . Thus, the flow on e is zero. $\square$

A famous unsolved conjecture of Tutte [19] states that every bridgeless graph has a nowhere-zero 5-flow. Seymour [15] has shown that every bridgeless graph has a nowhere-zero 6-flow.

Theorem 5.2.6. A graph has a nowhere-zero 2-flow if and only if each vertex has even degree.

Proof In a nowhere-zero 2-flow each edge has weight 1 and it follows that the indegree of each vertex equals its outdegree. Thus, each vertex has even degree. Conversely, if each vertex of a graph has even degree, then by Theorem 1.0.2 there is a decomposition of G into cycles. Clearly, any cycle has a nowhere-zero 2-flow. If we assign weights and orientations so that each cycle in the decomposition has a nowhere-zero 2-flow, then we obtain a nowhere-zero 2-flow in the graph. $\square$

Theorem 5.2.7. A 3-regular graph has a nowhere-zero 3-flow if and only if it is bipartite.

Proof Let G be a bipartite 3-regular graph and let A and B be the parts. By Theorem 1.0.3, G has a proper 3-edge colouring. Let F be the 1-factor corresponding to one of the colour classes. Each edge of F is oriented from A to B and given weight 2. The other edges are oriented from B to A and given weight 1. This is easily seen to be a nowhere-zero 3-flow.

Conversely, let G be a 3-regular graph with a nowhere-zero 3-flow. It is clear that each vertex is incident with an edge of weight 2 and two edges of weight 1. If we let A be the set of vertices incident with the tail of a weight 2 edge, and let B be the set of vertices incident with the head of a weight 2 edge, then each edge of weight 1 has its tail in B and its head in A . Thus, G is bipartite. $\square$

Theorem 5.2.8. A 3-regular graph has a nowhere-zero 4-flow if and only if it has a 1-factorisation.

Proof Let G be a 3-regular graph with a 1-factorisation $\{F_1, F_2, F_3\}$. Then each of $F_1 \cup F_3$ and $F_2 \cup F_3$ is a 2-factor. Any 2-factor has a nowhere-zero 2-flow. Let (D_1, f_1) be a 2-flow on $F_1 \cup F_3$ and let (D_2, f_2) be a 2-flow on $F_2 \cup F_3$.

Define a flow on G as follows. If e is an edge of F_1 , then e has orientation $D_1(e)$ and weight 1. If e is an edge of F_2 , then e has the orientation $D_2(e)$ and weight 2. If e is an edge of F_3 and $D_1(e) = D_2(e)$, then e has the orientation $D_1(e)$ and weight 3. If e is an edge of F_3 and $D_1(e) \neq D_2(e)$, then e has the orientation $D_2(e)$ and weight 1. It is routine to check that this produces a nowhere-zero 4-flow.

Conversely, let G be a 3-regular graph with a nowhere-zero 4-flow. It is easy to see (using parity arguments) that each vertex is incident with exactly one edge of weight 2. Thus, the edges of weight 2 form a 1-factor of G . The edges of weight 1 or 3 thus form a 2-factor F of G . It suffices to show that every cycle of F has even length, because then F has a 1-factorisation and this together with the 1-factor formed by the edges of weight 2 yields a 1-factorisation of G . Let C be an arbitrary cycle in F .

Consider the subgraph H of G induced by the vertices of C . A vertex of H has degree 3 in H if the incident edge of weight 2 is in H , and has degree 2 in H if the incident edge of weight 2 is not in H . Thus, H has an even number of vertices of degree 3, and so it suffices to show that there are an even number of vertices of degree 2 in H .

Consider the net flow out of $V(C)$, which is zero by Lemma 5.2.3. The net flow out of $V(C)$ is also equal to twice the number of edges of weight 2 leaving $V(C)$ minus twice the number of edges of weight 2 entering $V(C)$. These two numbers are thus equal and so there are an even number of vertices of degree 2 in H . Hence C has even length and G has a 1-factorisation. $\square$

Theorem 5.2.9. Let G be a plane bridgeless graph. Then G is k -face-colourable if and only if it has a nowhere-zero k -flow.

Proof First suppose G is a k -face colourable. We can assume that the colours are $0, 1, \dots, k-1$ and that the exterior face is assigned colour 0. We orient each edge e of G so that the colour c of the face on the left of e is larger than the colour c' of the face on the right of e , then we assign weight $c - c'$ to e . This produces a nowhere-zero k -flow. It is nowhere-zero because adjacent faces are distinct colours, and it is easy to see that if it is a flow, then it is a k -flow.

To see that it is a flow consider an arbitrary vertex v , and let $x_1, x_2, \dots, x_t$ be the colours, in clockwise order, of the faces incident with v . The edge separating the faces of colours x_i and x_{i+1} contributes $x_{i+1} - x_i$ to the flow into v (regardless of whether $x_1 > x_2$ or $x_2 > x_1$), and so the net flow into v is

$$(x_2 - x_1) + (x_3 - x_2) + \dots + (x_t - x_{t-1}) + (x_1 - x_t) = 0.$$

We have shown that if G is k -face-colourable, then it has a nowhere-zero k -flow.

Now suppose G has a nowhere-zero k -flow. Assign colour 0 to the exterior face. For each face F , the colour assigned to F is determined by travelling along a path in the plane from the exterior face to F , without passing through any vertex, and summing the flows of the edge crossings as follows. When an edge e is crossed and e is oriented left-to-right relative to the direction of travel, the flow on e is added to the sum. When an edge e is crossed and e is oriented right-to-left relative to the direction of travel, the flow on e is subtracted from the sum. The colour assigned to F is obtained

by reducing the sum modulo k . The colours of adjacent faces thus differ by the weight (which is non-zero and at most $k - 1$) of the flow on the edge that separates the faces.

We need to show that our colouring of the faces is well defined by showing that it is independent of the path taken from the unbounded face to F . It is clear that any path can be converted into any other path by going the opposite way around some vertices. So it suffices to show that going the opposite way around a vertex does not change the colouring. Observe that if we cross an edge and then (without going around any vertices) cross it again, then we are back in the same face and the contribution to the sum from the second crossing cancels out the first, because in one crossing the flow is left-to-right and in the other crossing the flow is right-to-left.

Suppose that going in a clockwise direction around a vertex v we have edges $e_1, e_2, \dots, e_t$ and for $i = 1, 2, \dots, t$ let F_i be the face next to v and lying between e_i and e_{i+1} , with F_t lying between e_t and e_1 . For the portion of a path travelling clockwise around v from F_i to F_j , so via $F_{i+1}, F_{i+2}, \dots, F_{j-1}$, the sum Σ_1 of the flows for the edges crossed is the net flow into v from the edges $e_{i+1}, e_{i+2}, \dots, e_j$. For the portion of a path travelling counterclockwise around v from F_i to F_j , so via $F_{i-1}, F_{i-2}, \dots, F_1, F_t, F_{t-1}, \dots, F_{j+1}$, the sum Σ_2 of the flows for the edges crossed is the net flow out of v from the edges $e_i, e_{i-1}, \dots, e_1, e_t, e_{t-1}, \dots, e_{j+1}$. Thus, $\Sigma_1 - \Sigma_2$ is the net flow into v , which is zero, and so $\Sigma_1 = \Sigma_2$. This shows that our colouring of the faces is well-defined, and we have thus shown that if G has a nowhere-zero k -flow, then it is k -face-colourable. $\square$

5.3 Two Theorems of Petersen

The following two theorems were proved by Petersen in 1891 [13]. The first, Theorem 5.3.1, is generally known as “Petersen’s Theorem”. Our proof of Theorem 5.3.1 uses Tutte’s 1-Factor Theorem, which was not proved until 1947.

Theorem 5.3.1. Every bridgeless 3-regular graph has a 1-factor.

Proof We use Tutte’s 1-Factor Theorem (Theorem 5.1.1) to prove the result. Let G be a bridgeless 3-regular graph, let $S \subset V(G)$, let $o(G - S)$ denote the number of components of odd order in $G - S$, let H be a component of odd order in $G - S$, and let m be the number of edges joining H to S . The sum of the degrees of the vertices of H is $2e(H)$ and is also $3n(H) - m$. Thus, since $n(H)$ is odd, m is also odd. Since G has no bridges, this means that $m \geq 3$.

Thus, the number of edges of G that join odd components of $G - S$ to S is at least $3o(G - S)$. Since each vertex of S is incident with at most 3 such edges, this implies that $|S| \geq o(G - S)$. Thus, by Tutte’s 1-Factor Theorem, G has a 1-factor. $\square$

Theorem 5.3.2. Every regular graph of positive even degree has a 2-factor, and every regular graph of even degree has a 2-factorisation.

Proof Let G be a regular graph of positive even degree $2k$ and let the vertices of G be $v_1, v_2, \dots, v_n$. By Theorem 5.2.6, G has a nowhere-zero 2-flow, and thus there is an orientation D of G in which the indegree of each vertex is equal to its outdegree. That is, each vertex of D has indegree k and

outdegree k . Define a bipartite graph B with parts $U = \{u_1, u_2, \dots, u_n\}$ and $W = \{w_1, w_2, \dots, w_n\}$ by joining u_i to w_j if and only if $v_i v_j$ is an edge in D .

Since the outdegree of each vertex of D is k , the degree in B of each vertex of U is k . Since the indegree of each vertex of D is k , the degree in B of each vertex of W is k . That is, B is k -regular. Thus, by Theorem 1.0.3, B is k -edge colourable. Let F be a 1-factor of B induced by one of the colour classes in a proper k -edge colouring of B . If we define a subgraph F' of G by letting $v_i v_j$ be an edge in F' if and only if $u_i w_j$ is an edge in F , then F' is a 2-factor of G .

The second part of the theorem, that every regular graph of even degree has a 2-factorisation, follows immediately from the first part of the theorem because removing the edges of a 2-factor from a regular graph of positive even degree results in a regular graph of even degree. So the process can be repeated until a 2-factorisation is obtained. Alternatively, the colour classes of the proper k -edge colouring of B define the 2-factors of the required 2-factorisation. $\square$

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